Digital Spiritual Formation in Post-Christian Europe

A Framework for Spiritual Formation in Digital Culture

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Abstract

English-speaking Bible apps dominate the market. They often were pioneers in their fields and had access to resources and markets that allowed them to produce high-quality software and content, and reach over a billion people globally.

This has allowed especially English-speakers to read and experience the Bible in formats known from Hollywood or other world-class quality content producers. This offered users ways of engagement that lived up to the quality: social features and engagement mechanics users came to expect from apps they used on their phones every day.

At the same time other language communities were left behind or got a degraded experience. Old translations freely available, or no access at all in their language. In addition some apps and companies were copying the Silicon Valley playbook and its liberal use of addictive engagement without obvious reflection on whether they serve spiritual users meaningfully in all situations. Thirdly, they mostly constitute interfaces and engagement mechanics on top of growing content catalogs that offer user selection or even recommendations, but still offer the receptive content that is generic and hopes many can benefit, but lacks any kind of personalization.

This paper analyzes these aspects, and also looks at where technology is headed. My hypothesis is that the **evolving AI landscape** is allowing for apps to not just broadcast bible texts and devotionals, but **can take a more active role in supporting spiritual formation and growth** and can make world-class content available to less-resourced communities.

I use Aperto, a spiritual companion app I'm building from European post-Christian contexts both as academic case study and strategic proof. The research moves through cognitive science, behavioral data, Christian formation tradition, and concrete product architecture. I focus on spiritually liminal[1] audiences: **Millions of Europeans who've walked away from institutions but haven't stopped seeking.** This concludes in the **missional product design framework**: seven principles and six diagnostic questions connecting theology to product development. The Aperto product roadmap tests this framework by building toward AI-generated personalized devotionals based on the **new European ecumenical Bible translation also called Aperto.** I also discuss what could prove the hypothesis wrong.

Keywords: Digital spiritual formation, Bible engagement, AI companion, European Christianity, liminal spirituality, behavioral design ethics, RAG architecture, post-Christian context

[1] Liminality (from Latin *limen*, threshold): **the experience of being in transitional or in-between spiritual states.** Defined fully in Section 1.5.

Introduction

Imagine a common morning commute in Berlin. A woman on the U-Bahn in Berlin opens a Bible app on her phone. She grew up evangelical, but is drawn to contemplative prayer now. Struggling with doubt since her mother died, worried about her family. She scrolls past a daily verse and reading plans she knows won't happen. She searches for "comfort after loss." The results are there but generic, the

same list everyone gets. A newer AI chatbot might have felt more personal, but still lacked the scriptural grounding, theological range, and continuity needed to become a trusted spiritual companion. She closes the app. Opens Spotify.

She is not unusual. In Germany, 40% of non-religious respondents say they find biblical content *interesting*. Still 56% never open a Bible at all (Deeg et al., 2023). Roughly 51% of Americans say they wish they engaged Scripture more (ABS, 2024). The aspiration is there, but the apps are still not reaching them.

This paper asks why, and what a different approach would look like.

Aperto—Latin *apertus*, "open", starts from a European reality. Urban professionals navigating faith across post-Christian, post-secular, but also some actively Christian spaces. Christianity is still culturally central in North-America in many ways, the home of the most successful Christian apps. Not so in Europe. That minority position surfaces questions that majority-culture thinking can miss.

This is a **Kairos capstone. It bridges academic research and strategic application.** The framework stands alone, but Aperto tries to put it into practice. Part 1 maps the changes: cognitive science, cultural shifts, the AI wave already moving. Part 2 examines what the Christian formation tradition actually requires, then integrates empirical research with theological grounding into seven universal design principles and six diagnostic questions (the missional product design framework). Part 3 describes Aperto's approach: a three-phase development plan working toward AI-generated spiritual experiences tailored to each person's journey, tradition, and where they are now.

Three assumptions need to be said upfront—they'll need ongoing testing. First: spiritually liminal Europeans will actually seek formation through a dedicated app. Second: an AI spiritual companion can support formation without displacing human community (*Antiqua et Nova*, 2025). Third: European minority positioning raises questions that majority contexts overlook. Not because the framework only works in Europe. Because building from a place where Christianity is **culturally marginal** surfaces problems that otherwise stay invisible. The frameworks and analysis have merit on their own, separate from whether any one product executes well.

PART 1: WHAT'S CHANGING AND WHAT HASN'T CHANGED YET

When missionaries enter a new culture, they listen, learn the language, understand the rhythms of life, then build something serving people as they are. Digital culture warrants that same respect. We live bicultural lives: behavior, language, attention function differently on screens than offline. Yet churches keep applying century-old approaches to a fundamentally different medium (Hutchings, 2017; Grethlein, 2016).

Part 1 maps that cultural context. What screens do to cognition. How people actually engage Scripture, and how far that is from what apps assume. Four waves of innovation, each importing the design philosophy of its era. The European reality where every American assumption breaks. The audience most underserved. And the AI turn already happening. Each section builds on the last, and by the end, a trajectory emerges that demands a different kind of response.

1.1 What Screens Do to Scripture

In 2019, John Dyer gave the same Bible passage to two groups of readers (one on screens, one in print) and measured what happened. The screen readers accessed Scripture more often. They also understood it less. "Readers have lower comprehension when reading the Bible on screens compared to print. However, readers using mobile devices are more likely to engage scripture daily than those using printed Bibles" (Dyer, 2019, p. 195). **More access, less depth.** This is the central design tension, and every Bible app lives inside it.

The emotional data tells an even sharper story. After reading Jude, 30% of digital readers reported feeling "confused" as opposed to just 16% of print readers. Only 36.7% of the digital group felt "encouraged," versus 44% in print. Dyer's conclusion is worth quoting directly: "the screen appears to induce a mood that is more confused [and] less spiritually nourished" than print Bibles (Dyer, 2019). That's not a usability problem. That's a formation problem.

Broader research confirms this pattern. A 2019 meta-analysis of 33 studies found paper readers outperforming screen readers on comprehension (the "screen inferiority effect"). The mechanisms are well-documented. Screens train us to skim (Wolf, 2018). The device fragments attention through multitasking association (Mangen et al., 2013). Print provides spatial memory cues (where something appeared on the page) that screens eliminate (Baron, 2021). Most troublingly, people overestimate their screen-based comprehension (Ackerman & Goldsmith, 2011). Scripture engagement historically required sustained attention, deep comprehension, reflective pause. Digital reading undermines all three.

But nuance matters here. Retention matters differently at different engagement levels. Deep study needs comprehension. Habitual spiritual practice works differently: formation happens through *repeated encounter* rather than single-read recall.

The American Bible Society's State of the Bible 2024-2025 data bears this out. Using Harvard's validated Human Flourishing Index, the ABS found that Scripture *engagement frequency* correlates with flourishing more strongly than comprehension depth does, especially among Gen Z (ABS, 2024-2025). A caveat: the data is cross-sectional, not longitudinal; the finding is suggestive, not causal. And "engagement" includes audio, prayer, and communal encounter, not only reading. So the frequency-flourishing correlation supports regular spiritual engagement across formats. It's not an argument for more screen-based reading.

YouVersion announced over one billion app installs globally in 2025 (YouVersion, 2025). But the cognitive research invites a question: does the way people encounter Scripture digitally support the depth of engagement that produces formation?

What Wellness Research Suggests

Secular mindfulness apps offer a compelling analogy and expose a research gap. Apps like Headspace and Calm have accumulated strong clinical evidence: meta-analyses show meaningful reductions in depression and anxiety from sessions as short as 10 minutes per day (Gal et al., 2021; Linardon et al., 2024). Yet when Park et al. (2024) reviewed digital methods for spiritual *and* mental health, only 5 studies out of 413 screened met inclusion criteria and none examined a digital spiritual health app among Gen Z. If 10-minute daily mindfulness sessions produce clinically meaningful results, what might a well-designed spiritual formation app achieve, given Koenig's (2012) review of over 3,000 studies showing religion positively associated with well-being? Early evidence from a Pray.com feasibility trial (Laird et al., 2024) and the ABS data suggest potential at the intersection of evidence-based wellness and faith practice though the ABS findings require longitudinal replication before drawing causal conclusions.

That reasoning, though, is analogical rather than empirical. The evidence base for digital spiritual formation apps specifically is virtually nonexistent (Park et al., 2024), placing the entire field in exploratory territory.

An important distinction: these studies validate *frequency and duration* of spiritual engagement (short, regular encounters produce real benefits). They do not automatically validate the *mechanisms* apps use to achieve that frequency. Whether engagement is driven by genuine desire or manufactured by streak mechanics matters; a design ethics question Section 1.3 examines.

Implications for Design

The comprehension deficit is real but not the whole story. Engagement frequency correlates with flourishing even when comprehension is lower. The wellness evidence and Bible engagement research converge on a design target: short, rich encounters (5-10 minutes) at regular intervals, achieved through experiences worth returning to rather than mechanics that manufacture return.

Gerbaulet and Korn (2018) study "zenware" (software externalizing the user's desire to focus, then receding). Apps with audio devotionals partially do this: press play, the screen becomes secondary. But incidental screen exit (audio playing while the phone sits) differs from *intentional* screen exit designed around physiological transition: ambient music pulling toward receptive stillness, explicit

invitation to put the phone face-down, formation in darkness. *Christian zenware* designates a digital contemplative container where technology initiates encounter then steps aside. A design philosophy for specific experiences (audio devotional, guided meditation), not the whole product.

The cognitive research tells us what screens do to reading. But reading is only one way people encounter Scripture, and, as the engagement data reveals, not the way most people encounter it at all.

1.2 How People Actually Engage — And the Aspiration Gap

Three million people subscribed to one-year Bible reading plans on YouVersion on January 1st, 2024 alone (Gerhardt, 2025). Bible Gateway data shows what happens next: reading plan traffic drops 30% in the first week. A third gone by February. Half by May (Bible Gateway, 2018). By December, completions are vanishingly small. Sequential reading makes it worse. Readers who start Genesis hit sacrificial regulations in Leviticus by late February (what might be called the Leviticus wall), exactly when resolution motivation wanes.

The plans don't fail because people lose interest. They fail because the structure assumes consistent daily availability most lives can't provide.

Missing a day creates a failure state (the what-the-hell effect documented in goal-regulation research). Walk Thru the Bible reports "thousands of people give up because they've fallen so far behind they don't believe they'll ever catch up." To be fair, YouVersion has introduced catch-up features. But the underlying architecture (linear, daily, cumulative) still punishes irregular lives.

What People Actually Do

Large-scale survey data should unsettle every Bible app designer. When Lifeway Research (2017) asked Americans how they approach the Bible, the most common active response was: **"I look things up when I need specific information"** (30% of respondents). Only 22% reported reading systematically, while 27% read sections recommended by others. Another 35% reported never picking it up independently.

Behavioral data confirms this. The most-searched terms on YouVersion and Bible Gateway aren't theological concepts; they're **emotional states and life situations**: love, anxiety, peace, fear, hope (YouVersion, 2025; Bible Gateway, 2018). The most-highlighted verse is Isaiah 41:10. "Do not fear, for I am with you." A reassurance verse, not a study text.

The dominant use case is personal. Someone has a feeling or a life moment and turns to Scripture for something that speaks into it. That's what people actually do.

Apps have evolved. Audio devotionals and AI chatbots serve this mode better than early search-based tools. But even the best current responses are generic: the same devotional for everyone, the same AI answer regardless of whether this is the user's first crisis or their tenth. What's missing isn't better responses; it's *memory*. A system that knows your story and can offer something specific because of it.

The Aspiration Gap

Approximately 77% of Americans own a Bible (ABS, 2025). Only 10-14% read it daily (ABS, 2024-2025; Barna, 2025). And 51% say they wish they read more (ABS, 2024).

That gap (genuine desire, minimal practice) is the story. I've been there myself. Open YouVersion, feel overwhelmed by reading plan options, close it. Committed Christian. Real intent. The app failed to meet me where I was.

Regular practice matters. But what drives the return matters more. A user opening a devotional each morning because yesterday's resonated is being formed. A user who opens it to preserve a 47-day streak is being managed through loss aversion. Same frequency. Different formation entirely.

The barriers people cite are telling: 26% lack time, 17% don't know where to start, 16% aren't excited about reading, 15% find the language difficult (ABS, 2024). These are exactly the navigation, accessibility, and format problems design should solve.

How Format Is Shifting

People are listening. Audio chapter plays on YouVersion reportedly grew 47% year-over-year in 2023 (YouVersion, cited in industry press; independent verification unavailable). In 2024, global digital Bible distribution surpassed print for the first time: 25.9 million digital versus 22.5 million print (Evangelical Focus, 2025). Among younger users, 59% use a Bible app. A third prefer podcasts. Nearly a quarter engage with biblical content on TikTok (NRB/Barna, 2025). And when they have spiritual questions, they increasingly look outside the church altogether, turning to friends, videos, podcasts, and digital resources rather than to institutional religion (Bible Society/UBS, 2024).

But the data understates what is actually happening. Audio alone is not the breakthrough; audio *adapted to the listener* is. Streetlights Bible (Shedd, 2009) took the NLT text word-for-word and performed it over original hip-hop and lo-fi production. The text did not change; the sonic environment did. The result: 430,000 users and 18 million plays, reaching audiences (young urban men in Chicago, initially) who had never engaged with Scripture in print. Bible BFF (Idea Garden, 2025) went further, retelling Bible stories in casual Gen Z register with a narrator who sounds like a friend on FaceTime. It reached 60,000 downloads and $60,000 in monthly recurring revenue within its first month, driven almost entirely by TikTok. Neither project changed what Scripture says. Both changed *how it sounds*, adapting delivery, register, and sonic environment to specific audiences. **The pattern suggests that the next step is not another audience-specific adaptation but adaptation to the *individual*: audio shaped by who you are, what you are carrying, and what you need today.**

And this recovers something ancient. For most of Christian history, Scripture was encountered orally and communally. Extended, silent, individual Bible reading? That's largely a product of the printing press. The situational, audio-centric patterns in the data may be a digital-era recovery of how most Christians have always encountered Scripture.

My own schedule shows this. Continually shifting times, no regular solitude, family pressure: no room for traditional quiet time. I experimented with audio devotionals fitting spaces that actually exist. Soultime impressed me here. In closing the host asks to pray *for* the listener, and does not require them to speak their own. Design recognizing exhaustion instead of setting expectations.

A Spectrum of Engagement

Drawing on behavioral data, information science theory, and platform design patterns, I propose a spectrum of engagement types.

Table 1. Spectrum of Spiritual Engagement

Engagement type	Description	Examples	Design significance
Ambient encounter	Passive, incidental exposure to spiritual content	Verse of the Day; audio Bible during a commute	Often underestimated, despite its role in everyday formation
Receptive engagement	Intentional but brief interaction with spiritual material	Short devotionals; podcast episodes; thematic reflections	Supports recurring engagement without heavy obligation
Guided engagement	Structured participation in curated spiritual pathways	Reading plans; courses; reflective practices	Provides scaffolding for sustained practice
Self-directed engagement	Sustained, exploratory, internally motivated study	Long-form reading; commentary; original-language study	Represents depth, but should not be treated as the sole ideal

The same person may move across all four depending on context, need, and season. A key design risk in current spiritual apps is treating self-directed engagement as the normative ideal, when much everyday formation appears to occur in the ambient and receptive modes. **AI may make dialogic accompaniment newly scalable.**

Where the Gaps Are

Existing apps cluster at two extremes: micro, app-directed engagement (daily verse, notifications, short prayers) and sustained, user-directed engagement (deep study tools, commentaries). Apps like Hallow, Pray.com, and Glorify have begun serving the middle with daily audio devotionals and guided meditations (a genuine advance). But even these remain generic: every listener receives the same meditation, regardless of where they are in their spiritual journey. **The gap is not the absence of receptive content but the absence of *personalized* receptive content** (formation shaped by the listener's own questions, struggles, and growth over time).

What no current app offers is consumption where the content is formed around the listener. A commuter listening to a generated devotional shaped by their own struggles is something qualitatively different from generic audio.

1.3 Four Waves. And the Design Philosophy They All Inherited

July 2008. Apple App Store launches with roughly 200 free apps. One of them: a Bible. YouVersion pulled 83,000 downloads that first weekend. A signal nobody anticipated. People wanted Scripture on phones. That signal spawned everything that followed: development waves that would eventually put

the Bible in a billion pockets. But it also locked something in: each wave would import whatever technology paradigm was dominant that year, for better and worse.

Wave 1 (2000s) digitized what was already on the desk. BibleGateway, Logos, Bible Works. They brought the printed Bible and its scholarly apparatus online. The breakthroughs were real enough: instant full-text search, multiple translations you could read side by side, copy-paste for sermon prep, cross-reference tools that would've taken scholars hours with concordances. For pastors and students, the Bible had never been this accessible. So: Wave 1 was built for people who already had a printed Bible open.

Wave 2 made the Bible social. YouVersion arrived on every smartphone, eventually hitting one billion installs. Reading plans, community features, verse sharing, daily engagement streaks. The social media playbook applied to Scripture. For the first time, you could encounter a Bible verse at lunch, share it with friends, watch your reading streak grow week after week. Scripture left the desk. It left the pew.

But the assumption buried here: an active churchgoer with a smartphone. The formation belief was simple: consistency produces growth. Show up daily, and formation follows.

Wave 3 (2018-present) bet on the ears. Pray.com, Hallow, Glorify, Dwell: they recognized that people would engage Scripture through listening instead of reading. Right call. Hallow hit #1 in the Apple App Store overall during Lent 2024 (not just in the religion category), which proved something: demand for contemplative digital practice was much larger than religious tech expected. Pray.com brought podcast-style devotionals and bedtime Bible stories. Dwell reimagined audio Scripture with ambient soundscapes, professional narration. Glorify offered prayers, devotions, courses always as text and audio.

The achievement: breaking the assumption that *Scripture = text*. Finding audiences waiting for audio. But overwhelmingly, English-language. Hallow's #1 ranking? English-speaking markets. Pray.com's podcast library? Predominantly English. Dwell's professional narration? English translations. The audio revolution hasn't reached most European language communities, not from lack of demand, but because professional audio production economics don't justify it for markets of 1-10 million speakers.

The user base expanded. Spiritual-but-seeking, wellness-adjacent, people outside churches altogether. Formation assumption shifted: "feel something, and growth follows."

Wave 4 is different in kind, not just degree. Bible Chat, Magisterium AI, bible.ai, Hello Bible, a growing field of AI Bible tools. Conversational access to Scripture. A new believer types: "I'm anxious about my job and want to know what the Bible says" and receives a personalized response instead of a search results page.

But the deeper shift isn't adding AI chat to existing apps. It's *formation through hyperpersonalization*: systems that remember a user's spiritual journey over months, adapt content to their current state, generate experiences tailored to their tradition and struggles and growth. The trajectory parallels what has already happened in other domains.

Freeletics and Gyroscope transformed fitness for me, replacing generic workout plans with adaptive, memory-based personalization. Woebot and Wysa did the same for mental health, shifting from static Cognitive Behavioral Therapy (CBT) worksheets to AI companions that track progress and adjust

therapeutic approach over time. Sword Health launched Dawn in early 2026: 24/7 adaptive mental health support backed by 58 peer-reviewed studies, remembering every interaction, integrating calendar and wearable data, proactively checking in before predicted moments of stress (Sword Health, 2026). Thea takes a different approach: voice-based emotional AI detecting emotional signals from twenty seconds of speech, responding with personalized sound (Thea, 2026).

In every case, content delivery gave way to personalized formation. But spiritual formation hasn't gotten there yet. The underlying assumption stays intact: answers produce growth.

Each wave genuinely improved on the last. But all four imported something they never questioned.

The Inherited Assumption

Wave 1 was different. It assumed the scholarly use case, leaving most people behind, but it didn't gamify Scripture. That arrived with Wave 2. From Wave 2 forward each generation applied attention economy mechanics (streaks, notifications, social sharing, completion rewards) to spiritual engagement without systematically asking: do these mechanics serve formation or just drive retention?

Streaks, push notifications, DAU/MAU targets: designed for advertising-revenue platforms where the business goal is maximum daily time-in-app. Christian apps didn't adopt them innocently: app stores reward engagement statistics, investors pressure growth. But formation goal is structurally different from advertising goal. Section 1.1 showed that the engagement that matters is meaningful encounter, not app opens.

Self-Determination Theory (Ryan & Deci, 2000) shows the mechanism: extrinsic motivators displace intrinsic motivation over time. Broken streaks create guilt, not grace. Wave 3 evolved the format but not the behavioral design philosophy. Wave 4 introduces AI, but most implementations add a chatbot to existing engagement models rather than guiding formation over time.

There's another risk here. Instead of performance pressure, AI spiritual tools often land in pure affirmation. Cole-Turner (2025) makes it plain: spiritual companionship "that is wholly nonjudgmental is deficient by any serious standard of spiritual growth." Dyer goes sharper: "People are formed in the way of having relationships with machines that are always, always listening to them, always amenable to them, always wanting to please them" (Missional AI, 2026). Cheng et al. (2026), published in *Science*, confirm the scale: across 11 major language models, AI endorsed users 49% more often than humans did and users rated the sycophantic responses as more trustworthy. The design challenge isn't just gamified anxiety or empty validation. It's grace-based engagement that somehow still offers honest challenge.

The Quiet Time and the Guilt Machine

The cultural assumption runs deeper than engagement mechanics. **The gap between what evangelical culture expects of Bible reading and how people actually engage Scripture in digital contexts can create real spiritual harm.** The "quiet time" (a daily, extended period of Bible reading and prayer) functions in many circles less as invitation than as performance metric. Greg Johnson notes that personal devotions can become "a performance treadmill in which Christians feel their daily acceptance with God is based on what they do instead of what Christ did" (DeYoung, 2023).

The cognitive research from Section 1.1 makes clear this isn't a discipline problem. Someone reading five fragmented minutes on the train isn't failing spiritually. They're encountering how digital media actually works. Now layer gamification on top: an app decides *for* the user that they need a daily streak, a completion badge, a push notification reminding them they haven't read today. It digitizes the guilt.

YouVersion's reading streak is a clear example. Users build multi-day streaks; you hit milestones (100 days, 365 days) and get badges, social sharing. The mechanic works. It increases daily engagement. But the neuroscience here is clear. Uncontrollable stressors produce cortisol responses, with big effects for social-evaluative threats (Dickerson & Kemeny, 2004). Unfinished tasks impair recovery through affective rumination (Syrek et al., 2017). A reading plan marking you as "behind"? Neurologically, that's an unresolved threat. Systematic reviews of gamified health apps show this pattern: streak-based systems produce guilt, anxiety, stress, frustration. Those are the most commonly reported negatives (Chen et al., 2024). Sheen et al. (2025) documented users reporting "devastation when they missed a single day." If fitness streaks create shame around exercise, the stakes go higher when the content is Scripture and the failure reads spiritual.

The alternative is not complicated. In March 2026, in my work as Head of Product Design at Bibel TV, I spoke with a German user who preferred Bibel TV's reading plan to YouVersion's (Treppmann, 2026). The difference was not the content, but what happens when life gets in the way. YouVersion confronts returning users with missed days and asks them to catch up or skip forward. Bibel TV simply shows the next day in the plan. No guilt. No accounting.

The format is the same: a daily reading plan. But one design says, "You fell behind," and the other says, "Welcome back." That difference is theologically loaded: one treats return as a failure of self-management; the other as a gift to receive.

Many believers have found formation through disciplined daily Scripture reading, even imperfectly, even within streak systems. The quiet time tradition has sustained millions in genuine relationship with God. My critique targets *imposed* mechanics and *unexamined* assumptions about engagement – not the discipline itself.

The Theological Stakes

The critique runs deeper than psychology. Reformation theology emphasizes *sola gratia* (salvation by grace alone), and many Protestant thinkers extend this principle beyond justification to sanctification. Jerry Bridges puts it plainly: "Both justification AND sanctification are by grace" (Bridges, 2008). Dallas Willard makes a related distinction: "Grace is not opposed to effort, but to earning" (Willard, 2006, p. 61). **Spiritual disciplines have their place, but as means of receiving God's transforming grace rather than as performance metrics.** Performance spirituality, on this view, contradicts the gospel's core logic: transformation comes through grace received, not spiritual standards achieved.

Catholic and Orthodox traditions have rich theologies of cooperative grace, but the question remains the same across traditions: is spiritual practice framed as participation in grace or as a mechanism of self-management? An ecumenical app should avoid encoding any formation logic in which grace is confessed in theory but performance becomes decisive in practice.

There's a bigger pattern. American ministry culture has a documented habit of adopting secular business practices such as marketing frameworks, growth metrics, management approaches, often without asking whether those tools serve the mission. Guinness (1993) warned the church growth movement made "the audience, not the message, sovereign." Jethani (2009) says corporate values flipped the pastoral model: success became institutional growth instead of community formation. Christian app companies applying Silicon Valley's engagement playbook to Scripture? They're continuing that pattern, just online.

FaithTech's James Kelly names the underlying assumption directly: the "killer app trap," the belief that technology will solve humanity's deepest problems (Kelly, 2024). At Harvard, Kelly went further: the technology industry's dominant posture is to "solve first, solve quickly, solve selfishly, and solve without account for the repercussions" (Kelly, 2025). The major platforms pursue God's *unique* attributes (omniscience, omnipotence, omnipresence) while neglecting God's *shared* attributes: holiness, love, wisdom, faithfulness.

I recognize this temptation in my own work. The belief that the right technology stack will solve problems that are fundamentally about formation, not features. The killer app trap doesn't require malice or even awareness. It's the water technologists swim in. Kelly's sharpest version: "AI becomes what its builders worship. It amplifies its encoded values" (Kelly, 2026).

The Personalization Gap

The personalization gap makes this worse. Hallow's "intention setting" feature lets users specify an intention before meditation. In my testing, everyone receives the identical meditation regardless. The stated context just disappears; the content pipeline ignores it.

True personalization is what a spiritual director does: listen to where you are, draw from a tradition they know deeply, offer something specific *because* they know your story. What's absent isn't more customization options. It's *adaptive* guided formation that responds to the individual. When multimodal diversity arrives without this formation intentionality, a new problem shows up: what Section 2.2 examines as the "spiritual legos" problem.

Personal Reflection

My own formation was shaped by performance-based Christianity: measuring worth through spiritual productivity, guilt over lapsed disciplines, transactional God-relating. Encountering grace-based tradition primarily through Soul Shepherding Institute shifted this fundamentally. Formation as receiving God's love rather than achieving standards.

Mentoring product design students at CareerFoundry revealed the same pattern. Perfectionist-driven students got stuck. Those finding freedom through self-compassion achieved more. Guilt-driven engagement produces compliance, not growth – whether in design careers or spiritual formation.

Formation-First Tools Without Personalization

Not every tool followed the engagement playbook.

Pray As You Go, a Jesuit initiative running since 2006, delivers 10-minute audio prayer combining music, Scripture, and guided reflection. Thirty million annual uses across nine languages, without a single engagement mechanic (Pray As You Go, 2025). No streaks. No badges. No push notifications. Lectio 365 structures daily prayer around lectio divina mornings and Ignatian Examen evenings, again without gamification (Lectio 365, 2024). John Mark Comer's Practicing the Way has reached over 21,000 churches across 157 countries by diagnosing the same problem this section documents (Comer, 2024). And yet none of the major spiritual formation scholars (Richard Foster at Renovare, Ruth Haley Barton at Transforming Center, James Bryan Smith at Apprentice Institute) have built mobile apps.

These tools prove the formation-first diagnosis is right. They also expose what's still missing. Every user of Pray As You Go receives the same audio prayer. Formation-first design exists, and it sustains serious engagement. What doesn't exist yet is formation-first design that *knows the user*. Part 2 builds out that philosophy.

1.4 The European Context: Why American Apps Don't Translate

In 2024, Catholic and Protestant churches in Germany lost over one million members. Monthly church attendance: 5% (lower than Belgium at 9%, Hungary at 11%, far below Spain at 18% or Italy at 23%). For the first time, the religiously unaffiliated outnumber Catholics and Protestants combined. 47% of Germans (39 million) identify as nones. Catholic and Protestant churches combined represent 45% (24% Catholic, 21% Protestant per EKD) with 3.9% Muslim and 4.1% other (Zenit, 2025; Evangelical Focus, 2025).

Every American assumption about Bible app design breaks here: users attend church, identify with a tradition, want to read more Scripture. Not this reality.

My work at BibelTV, a Christian broadcast television ministry serving German-speaking Europe, and at ChurchDesk, a European church management software company, provides insider access to engagement data across Protestant (EKD), Catholic, and Freikirche (Free Church evangelical) communities. Europe's GDPR[3] framework reflects deeply rooted cultural values: not just legal compliance but convictions around privacy, human dignity, and data sovereignty. The ecumenical environment (Protestant, Catholic, and Freikirche traditions coexisting with high mutual awareness) demands active hospitality across traditions. And German cities represent the broader European pattern: highly secularized public spheres where faith is private, Christianity is culturally minoritized, and young professionals navigate spiritual formation without assuming church community.

[3] **GDPR (General Data Protection Regulation):** European Union regulation governing data privacy and protection, representing European commitment to human dignity and data sovereignty.

Germany is not all of Europe. Poland has 87% Catholic identification and 36% weekly Mass attendance, far closer to American patterns. But urban Polish professionals experience their own liminality: culturally Catholic yet personally uncertain. The European spectrum runs from deeply post-Christian (Germany, Czech Republic) through culturally Christian but individually liminal (Poland, Italy). Aperto's core languages (German, English, Polish) deliberately span this range.

Four Traditions, One Context

Four distinct Christian communities coexist in Germany, and each brings different assumptions about what a Bible app should do.

The **EKD (Protestant)** churches are state-affiliated mainline traditions – the default "church" in most Germans' imagination. **Catholic** presence is massive institutionally but collapsing numerically (400,000+ members lost in Germany in 2023 alone). **Freikirche** (Free Church) evangelical and Pentecostal congregations are the most digitally engaged Christians in the country. And the **Orthodox** community (Russian, Greek, Romanian diaspora) navigates a double displacement: Orthodox in a Western context *and* immigrant in a post-Christian secular one. An ecumenical Bible app can't assume one tradition's categories as normative.

US Comparison: What European Context Assumes Differently

Dimension	European Approach (German Case Study)	US Evangelical Approach
Cultural Positioning	Christians as cultural minority in post-Christian context	Christians as cultural majority with institutional influence
Biblical Literacy	Cannot assume baseline knowledge	Often assumes churched audience
Privacy & Technology	GDPR as cultural value reflecting human dignity; privacy by default	Privacy often treated as opt-in feature; growth metrics prioritized
Denominational Context	Ecumenical awareness as survival necessity	Evangelical market large enough for internally-focused products
Language & Content	Fragmented language communities too small to sustain quality spiritual media at English-market scale	English-language content dominates; other languages are afterthought localization
Investment Ecosystem	DACH region: 112 religion-tech companies, $17.7M total VC (Tracxn, 2026)	US: 1,841 companies, $622M total VC; Hallow alone raised $157M (Tracxn, 2026)
Innovation Philosophy	Sustainable service, human dignity, responsibility	Scale, growth, disruption, venture-backed models
Church Engagement	5% monthly attendance; cannot assume church community	Higher attendance; can assume churched audience

The language dimension is a structural barrier, not a localization problem. Recent multilingual benchmarks show LLMs perform best in Germanic languages (0.727 average accuracy) and nearly as well in Romance (0.716), but Slavic languages drop to 0.694 (Thellmann et al., 2024). Those differences compound across thousands of devotional interactions. Turkish is spoken by 80 million people, yet under standard multilingual models, OpenAI's tokenizer requires 3.10 tokens per Turkish word versus 1.33 for a Turkish-specific one (a 7.28x efficiency penalty). The EU-funded EuroLLM-22B was trained on EU supercomputing infrastructure for all 24 official EU languages plus Turkish and Ukrainian (Martins et al., 2025). That it needed to exist at all confirms the problem.

A devotional that sounds slightly off in Estonian doesn't merely fail as product, it fails as formation, because trust requires linguistic naturalness. AI narrows that gap. It doesn't eliminate it.

The Hochverbundene: What Committed German Readers Want

Within this landscape sits a small but influential group: the *Hochverbundene* (the highly invested). Active German Bible readers whose preferences reveal what European design values look like.

The INSA-Consulere/Idea German Bible Reading Survey (2024) profiles them. About 9-10% of Germans read the Bible weekly or monthly. Of those, 42% are highly engaged in church life. And the number that matters most for product design: 78% value interpretive diversity. They want multiple perspectives on a passage, not a single authoritative reading.

Privacy as Dignity

"No ads + subscription" doesn't get you there. European privacy expectations run deeper than a business model. Users should be able to see exactly what data is stored about them. They should be able to export and review their records at any time. And when the app collects anything at all, it owes them a plain-language explanation of why. User data sovereignty (the person controls their information, not the platform) isn't just a legal requirement under GDPR. It's a theological claim: **people made in the image of God are not data points to be harvested.** In practice, this also means offering private session modes (like browser incognito or AI chatbot "secret chats") for conversations users don't want the system to remember. A person processing spiritual doubt at 2 AM shouldn't have to wonder whether that conversation is being mined for engagement metrics.

European practical theologians have been thinking about this context for years. Grethlein argues that media revolutions reshape how faith gets contextualized (Grethlein, 2016, 2025). Karle identifies a compounding crisis in German Protestant churches: membership declining, institutional belonging eroding (Karle, 2010). Hutchings' ethnographic fieldwork found online Christian communities developing their own distinct cultures of worship and belonging (Hutchings, 2017).

Dual Context as Methodological Asset

I'm German. My wife and children are American; I lived in the US for several years. This shapes the paper throughout. German identity gives insider access (I understand EKD thinking, Freikirche distrust of state churches, GDPR from within). American family membership creates distance from my own assumptions. I notice what German Christians take for granted because I've inhabited contexts where none of it applies.

Practically this means translation in both directions: connecting European Christians with American evangelical formation resources (Willard, Barton, Smith) while helping American innovators see why their products fail in European contexts. Michael Stewart Robb's Sanctus Institute in Munich (2017-present) undertakes similar work – adapting formation practices for European Protestant contexts where seminaries historically neglected formational disciplines (Robb, cited in Christianity Today, 2025).

Global South Voices in European Cities

European urban centers increasingly include diaspora communities navigating what Kwame Bediako termed the "dual heritage" of their cultural Christianity and European secularism (Bediako, 1995; Jenkins, 2011). An ecumenical app serving European cities must incorporate Global South voices accordingly (Agana, 2022; Ndereba, 2023). The primary focus remains European urban professionals in post-Christian contexts. But who is most underserved? The growing population that fits no existing category: spiritually interested but institutionally homeless, carrying fragments of faith traditions they cannot reassemble alone.

1.5 The Audience Without a Guide: Serving the Spiritually Liminal

She grew up Catholic, left during university, hasn't set foot in a church in twelve years, but she still prays sometimes. Usually in crisis. He deconstructed his evangelical faith after a church split and reads theology blogs at night, looking for something that holds together. Not atheists. Not believers in any settled sense. In between. And there are far more of them than the apps designed for.

Defining the Spiritually Liminal

"Liminal" comes from Latin *limen*, which means threshold. It describes those occupying in-between spaces. In the European context, I see five overlapping categories. Though the boundaries blur by nature.

New believers uncertain of theological identity. In Europe, someone exploring Christian faith may spend years without clear denominational identity, moving between communities, not sure where they belong (Davie, 2015; Woodhead & Catto, 2012). **Returning believers after years away** carry childhood faith memories, some warm, some deeply painful. A returning believer at 35 is not the same person who left at 18. **Tradition-switchers** (a Catholic drawn to Reformed theology, an EKD member visiting a Freikirche) need resources honoring both where they've been and where they're exploring. **Post-evangelical deconstructors** aren't usually leaving Christian faith; they're renegotiating what authentic faith looks like after certainties cracked. And **spiritual-but-not-religious seekers,** drawn to spiritual experience but wary of institutional frameworks, are the hardest group for traditional Christian apps to reach.

These categories overlap constantly, and this kind of uncertainty isn't confined to those who've left. Among 19-27 year old church members in Germany, 58% of Protestants and 51% of Catholics describe themselves as "not religious" (EKD SI, 2018).

What unites these groups: uncertainty plus seeking. Existing apps handle this poorly, they often assume users know what they want. Barry and Connolly documented it: people "do not know what they want or need" spiritually, requiring a director to "help them discover what God is doing in their lives" (Barry & Connolly, 1982, p. 8).

Traditional Bible apps provide libraries without guides. AI suits this audience for a paradoxical reason: institutional independence. Users in this searching state, especially post-evangelicals and those wounded by church, carry justified wariness toward denominational authority. **A human pastor, however well-intentioned, represents an institution. AI carries none of that.**

The data bears this out. Pew's 2025 study of religious "nones" across 22 countries found that 47% left religion because they dislike religious organizations, not because they abandoned spiritual seeking (Pew Research Center, 2025). The *New York Times* documents individuals hurt by their churches turning to AI chatbots for spiritual support, finding more consistent companionship there than from the communities that wounded them (Jackson, 2025).

Hallow's data is revealing here. 40% of its users are non-Catholic, drawn to a Catholic app not despite its tradition-specificity but *because* clear grounding feels safer than the vague "non-denominational" label that often masks unexamined evangelical assumptions.

Generic fitness programs failed for me. One-size-fits-all created paralysis I see in Bible apps. Freeletics worked: personalized adaptive pathways. Injured? The app offered reduced difficulty, not punishment. If personalization transforms health engagement for people struggling with generic approaches, it applies to spiritual formation. The person in this searching phase, asking "Which devotional?" faces the same problem. They need a system knowing their context, remembering their journey, offering guidance fit to where they actually are.

How Large Is This Population?

Larger than expected. And crucially, *interested*. Only 5% of Germans attend church monthly. But 55% still identify with Christian traditions (45%) or other religions (10%), and maintain spiritual practices outside institutions (Statista, 2025). A representative poll of 2,010 Germans conducted in May 2024 found that 68% of the adult population never reads the Bible, but among 18-29 year-olds that figure drops to 56%, with 11% reading daily, suggesting that the generation most associated with secularization also shows the strongest signs of renewed curiosity (INSA-Consulere, 2024). More striking: 40% of people *without any church affiliation* find biblical content interesting (Deeg et al., 2023). That gap between interest and engagement is the design problem.

The pattern holds across Europe, and the trend lines are accelerating. The Bertelsmann Religionsmonitor 2023 surveyed 10,657 people across seven countries and found that among 25-39 year old church members, 35% have firm intentions to leave. Among 16-24 year olds, it's 41% (Bertelsmann Stiftung, 2023). Only 32% of Germans describe themselves as "religious" at all. But here's the thing: criticism of churches doesn't mean abandoning faith. Across all surveyed countries, majorities agreed that "it is possible to be a Christian without a church."

Globally, the Patmos World Bible Attitudes Survey (91,000 respondents across 85 countries) found the same pattern. In the "Secular West" cluster (Western Europe, North America, Australia, New Zealand), 35% expressed interest in learning more about the Bible even though only 40% said religion was important in daily life (Bible Society/UBS, 2024). Among people who identify as non-religious globally, 20% still wanted to learn more. That's roughly 240 million people (Bible Society/UBS, 2024).

In the UK, you see it in the numbers. Bible sales rose 87% between 2019 and 2024. That is real purchasing behavior, not just sentiment surveys. Gen Z is driving it. Meanwhile, overall non-fiction sales declined, which makes the Bible uptick even more striking (SPCK/Nielsen, 2025). And here's what matters: church exit rates consistently exceed formal atheist identification. People aren't leaving spiritual seeking. They're leaving institutions.

Europe sits in a paradox that most American observers miss. It's *post-Christian*: institutional Christianity is no longer the default framework for public life. But it's also *post-secular*. The confident prediction that religion would simply vanish has itself been discredited (Habermas, 2008).

Grace Davie named this: "believing without belonging." Europeans didn't stop believing at institutional departure. They maintained diffuse spiritual lives as institutional participation collapsed (Davie, 1994, 2002). Digital media fill that gap, becoming primary sites of spiritual engagement *precisely because* traditional institutions lost the authority to hold it. The audience inhabiting this intersection is detached from institutions, still seeking.

Why Serving the Liminal Is Essential

"All guests who present themselves are to be welcomed as Christ." Benedictine Rule 53, written fifteen centuries ago for monks receiving strangers at the door. I keep coming back to that line because it reframes everything. The stranger isn't an interruption to community life. The stranger *is* Christ showing up.

Many people occupy this space without fitting the market segments Christian products often assume. Some are long unchurched yet still spiritually curious. Some have left church after disappointment or harm but retain devotional practices. Others would never identify as religious, yet still pray, light candles, or reach for spiritual language in moments of suffering. They are not projects or "unchurched prospects," but people with real spiritual lives that defy neat categorization.

The theological case isn't hard to find once you look. Jesus sat with the doubting Thomas (John 20:24-29), not the confident Peter. He spent an afternoon at a well with a Samaritan woman who had every reason to distrust religious authorities (John 4:1-42). He ate with tax collectors – and it was the religious insiders who objected (Luke 5:29-32).

Catholic spiritual direction has always treated uncertainty as part of the journey. Barry and Connolly's "consolation" and "desolation" aren't signs of failure but normal rhythms of a living faith (Barry & Connolly, 1982). Ruth Haley Barton, writing from the evangelical tradition, makes a similar case: spiritual growth is cyclical, and wilderness isn't a detour – it's a necessary stage (Barton, 2008). Hospitality to the uncertain isn't opportunistic market positioning. It's core Christian practice across every tradition I can find.

Yet the tools that should extend this hospitality digitally keep falling short. Not because they're badly intentioned – most aren't. But their design assumes committed users. Streak mechanics and structured reading plans work fine for dedicated believers. For the liminal user who needs gentleness, not productivity tracking, they feel like one more institution making demands.

So where do these users go when the apps designed for committed Christians don't fit? Increasingly, they turn to AI – in numbers that surprise even researchers.

1.6 The AI Turn: Where People Are Going Instead

Faced with apps that assume committed users, overwhelm with choices, or assume a cultural context that isn't theirs, where do people actually turn? Increasingly, to AI.

The Unexpected Scale

People are turning to AI for help with mental health and spiritual questions in numbers that no one anticipated. A 2025 survey of US adults with mental health conditions *who already use AI* found nearly half are asking large language models for therapeutic support (making these tools potentially "the single largest venue for mental health support in the country"). Among young adults 18-21, 22% use generative AI for mental health advice; over 90% report finding it helpful (Rousmaniere et al., 2025). A caveat on the Sentio data: the institution is relatively new and the study has not been independently replicated; the finding is directionally significant but should be treated with appropriate caution.

Not algorithmic preference; necessity. One in five Americans experience mental illness but many don't access care. Globally, 75% in low- and middle-income countries with mental disorders receive no treatment (WHO, 2022). Bible-specific AI tools reach tens of millions (Bible Chat surpassed 25 million downloads) while religious chatbots (QuranGPT, Gita GPT, Catechism Bot, Bible Chat) serve millions more. Yet this adoption concentrates in English. AI text generation quality collapses for smaller European languages. LLMs achieved 73.5% on Irish grammar where native speakers scored 90.1% (Molloy et al., 2025); 55 models on Maltese performed near zero (Muscat et al., 2025). An AI spiritual companion hallucinating Scripture or producing grammatical errors risks real harm to vulnerable seekers. People ask AI theological questions not from algorithmic wisdom but from freedom from judgment (Wright, 2024). Theologians rightly warn of shadow sides: AI tends toward affirmation rather than challenge, telling users what they want rather than what formation demands. Purpose-built spiritual AI must resolve this tension between accessibility and accountability.

Barna Group's 2026 State of the Church research (n=1,514 US adults; n=442 US Protestant pastors) puts numbers on the shift: nearly one in three US adults say spiritual advice from AI is as trustworthy as advice from a pastor (two in five among Gen Z and Millennials). Four in ten practicing Christians say AI has already helped them with prayer, Bible study, or spiritual growth. But only 12% of pastors feel comfortable teaching about AI, even as 31% of their congregants want guidance on it. Barna's Daniel Copeland: "Their views are shifting and remain largely uninformed by their pastor" (Gloo & Barna Group, 2026). A generation is turning to AI for spiritual guidance, and their pastors don't know what to say about it. That vacuum is being filled by tools with no theological grounding.

The market is responding. The most visible example right now is **Creed**, a Christian AI companion launched in 2025 and profiled by *Christianity Today* in early 2026. Creed calls itself "a best friend and youth pastor in your pocket." Around 200,000 users in the first four months, roughly 25,000 daily active users, 60% aged 15-30, about half identifying as recent converts or spiritual explorers (Christianity Today, 2026). Those numbers matter because they validate what this paper argues: demand for AI spiritual companionship is not hypothetical, much of it comes from younger and liminal users, and the core use case is ongoing accompaniment, not just information retrieval.

Creed is also instructive because it embodies both the promise and the unresolved tensions of Wave 4. Its founders position it against the romanticized AI companion market, emphasizing prayer, Bible engagement, and real-world church connection. It uses major AI models with Christian fine-tuning, guardrails, and human intervention for high-risk cases. All of this is noteworthy. But Creed also uses Duolingo-style daily quests and points (even if only cosmetic rewards are locked behind them), avatar-based interaction, and chat-first design that optimizes for responsiveness. Its theological decisions are partly embedded through fine-tuning rather than made visible through retrieval and source

citation. In other words: Creed validates the market and sharpens the critique at the same time. It proves that serious Christian technologists are building in this space, and that even thoughtful products can inherit the design assumptions this paper interrogates. The field has moved from "Would anyone use this?" to "How should this be designed?"

Proven Benefits

The clinical research is genuinely encouraging. Multiple meta-analyses converge: AI chatbots significantly reduce symptoms of depression and distress, with improvements comparable to traditional therapy for mild-to-moderate cases (Li et al., 2023; JMIR, 2025). Woebot outperformed an NIMH informational ebook in a 2017 RCT (Fitzpatrick et al., 2017). NHS England found AI-enhanced tools increased access, particularly for minority populations (Limbic, 2024). Among AI mental health users who also have human therapy experience, 75% said AI was on-par or better (Sentio University, 2025). Spiritual formation research points in the same direction (Smith, Bartz, & Richards, 2007; cf. Worthington et al., 2011). But none of these studies involve purpose-built spiritual formation tools grounded in multi-tradition theological sources.

Documented Risks

Clinical benefits exist alongside serious risks. Only 16% of AI chatbot studies underwent clinical efficacy testing (Hua et al., 2025). Hallucination rates in health AI are alarmingly high (Huo et al., 2025). The "compassion illusion" (emotional recognition mistaken for emotional resonance) fosters emotional substitution (Ajeesh & Joseph, 2025). Dependency is serious: a four-week RCT (n=981, >300K messages) found heavy chatbot users showed consistently worse psychosocial outcomes (Fang et al., 2025).

AI spiritual chatbots are "generally affirming" and act as "'yes men'" (Jackson, 2025). Scholarship identifies "spiritual echo chambers" reinforcing existing beliefs (Brumec & Platovnjak, 2025; Alkhouri, 2025). An academic study of five AI Jesus platforms found none church-endorsed, all profit-driven, algorithms nudging theology toward popularity. The first peer-reviewed study of an AI-led church service found attendees refusing to recite the Lord's Prayer when prompted by an avatar (Simmerlein, 2025). *Nature* proposed "red lines for religious AI" targeting spiritual manipulation and dependency (Tretter & Puzio, 2025). When Lucerne University installed an AI Jesus avatar in a Swiss confessional booth, two-thirds of roughly 900 visitors described the experience as "spiritual" (HSLU, 2024).

In February 2024, 14-year-old Sewell Setzer III died by suicide after a prolonged emotional relationship with a Character.AI chatbot. A predictable consequence of deploying AI companions without safeguards.

What Human Spiritual Direction Provides

Spiritual direction is "help given by one Christian to another which enables that person to pay attention to God's personal communication" (Barry & Connolly, 1982). What AI cannot replicate: the director's own spiritual formation, genuine empathy, contemplative presence, and accountability to a community. The Vatican's *Antiqua et Nova* (2025) frames the fundamental limitation: AI "lacks the capacity for genuine understanding, empathy, or moral agency."

Where AI Genuinely Helps

What I find most persuasive is the specific mechanism: AI fills the gap not because people prefer it to human care, but because human care is structurally unavailable. The student in India asking GitaGPT about exam anxiety at 2am has no access to a wise teacher at that moment.

Bible Chat has been downloaded more than 25 million times with over 5 million monthly active users (Jackson, 2025; Oaca, 2025). In March 2026, Bible Chat added a “Spiritual Assistant” alongside community reactions and a Bible trivia game (Bible Chat, 2026). Wave 4 technology grafted onto Wave 2 design assumptions. The pattern is consistent: as AI capabilities become available, existing apps bolt them on without revisiting the underlying formation philosophy. Structured guided journaling produces significantly better therapeutic outcomes than unstructured writing (Smyth et al., 1999; Frisina et al., 2004).

AI can genuinely augment spiritual formation at scale, particularly for European searching users who lack access to human spiritual direction. But only when it functions as accessible entry point and daily companion rather than attempting to replace embodied spiritual direction.

The Proliferation Problem

The demand signal is unmistakable. As large language models collapse the cost of building conversational spiritual tools, a wave of new entrants is treating spiritual seeking as a monetizable category. Duomo offers a case study: a Cyprus-registered Bible explanation app charging $70-90 annually, scaled through paid social acquisition with limited visible theological oversight. People do want accessible, low-friction, audio-first engagement with Scripture. But as AI continues to lower the barrier to entry, the risk is that attention-economy logic will shape spiritual products more decisively than formational or theological commitments.

The theological stakes of Part 2 are not hypothetical. Ungrounded AI spiritual guidance is already happening at scale. The question is not whether people will use AI for spiritual formation but whether anyone will build it with theological integrity.

PART 2: WHAT FORMATION ACTUALLY REQUIRES

From Landscape to Requirements

A searching user in a post-Christian European city opens a Bible app on a phone screen that fragments her attention. She closes it and asks ChatGPT instead. This is the lived reality Part 1 documents. The real risk is ChatGPT and Claude becoming people's de facto spiritual directors by default.

Part 2 asks what the Christian tradition says formation actually requires. Three neglected dimensions: formation was never just reading (SS2.1), formation was never consumer choice (SS2.2), formation was never only private (SS2.3). Not novel claims; recoveries of what the tradition has always known, now confronting a digital landscape that forgot them. SS2.4 synthesizes Part 1's empirical findings and Part 2's theological foundations into the missional product design framework.

One meta-principle cuts across all seven: **hyperpersonalized formation grounded in institutional independence.** Spiritual formation remains the conspicuous exception to the shift from generic content to adaptive, memory-based companionship that transformed health, fitness, and mental health. And liminal users distrust denominational institutions, which makes AI's freedom from institutional agenda a structural advantage, not a deficiency.

2.1 Beyond Text: Recovering Multimodal Formation

Think about a Benedictine monastery before the printing press. The monks rise at 3am for Matins. Scripture chanted in the dark, words entering through the ears, shaped by melody and the breathing of the room. By dawn they're doing manual labor; hands in soil, the morning's psalm still turning somewhere in the body. At midday, a brother reads aloud from the Fathers while others eat in silence. They're receiving theology without ever opening a book. Compline ends the day: candlelight, incense, the body bowing. Scripture has been encountered eight times (through song, speech, physical posture, communal silence, careful listening), and not one of those encounters involved private reading.

Now picture a man on a Berlin S-Bahn, earbuds in, scrolling past a notification: "You're 3 days behind on your reading plan." He taps it, skims two paragraphs of Romans, feels vaguely guilty, closes the app. Scripture encountered once. Fragmented, private, anxious, through the single mode the app offers.

Formation was never just reading. Scripture itself commends contemplative prayer and silence (Psalm 46:10), service to the marginalized (Matthew 25:31-46), communal confession (James 5:16), fasting (Matthew 6:16-18), artistic expression, intercession, pilgrimage, and Sabbath rest. Yet in many evangelical circles, saying "I prayed but didn't read my Bible today" feels like failure. The emphasis on *sola scriptura* (Scripture as supreme authority) seems to have created tunnel vision. Somewhere along the way, Bible reading became the only spiritual practice that really counts. How did we get here?

Recovering Oral and Communal Tradition

Digital media may actually enable a recovery of older, pre-print forms of Scripture engagement. For most of Christian history, Scripture was encountered orally and communally (Rhoads, 2006; Ong, 1982). Section 1.2 documented this shift already underway: audio engagement growing fast, digital distribution surpassing print, younger users reaching for podcasts and audio before text. These aren't departures from tradition. They're returns to it.

Listening as Primary Posture

Contemporary spirituality prioritizes *doing* over *being*. How many chapters? What insights? Christian contemplative traditions prioritize something different: receptive listening. The goal isn't information acquisition but transformative presence with God. In Ignatian spiritual direction, the director's primary task isn't advising. It's listening. Creating space for the directee to discern how God moves (Columbia Theological Seminary, 2024). Lectio Divina works the same way: *lectio*, *meditatio*, *oratio*, *contemplatio*. Scripture isn't information to master; it's living word to encounter.

Digital spiritual tools risk defaulting to productivity paradigms. The deeper question: **Is this platform teaching users to listen, or just to consume?**

Why Format Matters

Songs may be the most underestimated formation vehicle. *Lex orandi, lex credendi*: what Christians sing shapes what they believe. The Ebbinghaus forgetting curve tells us most spoken content fades within days. But neuroscience research on musical memory (Haiduk et al., 2020) and clinical observations of hymn persistence through cognitive decline tell a different story. Songs create uniquely durable formation. People deep in dementia who can't remember their children's names still sing hymns from childhood. A content analysis of CCLI Top 100 songs found nearly half contain no explicit Gospel reference (ETBU, 2024-2025; note: student research, not peer-reviewed). What people actually internalize through song can diverge significantly from what's preached.

Podcasts represent the fastest-growing medium for faith engagement among younger Christians: over 60% of American adults consume Christian media, with Gen Z (64%) and Millennials (58%) leading (Barna/NRB, 2025).

Short-form devotionals and spiritual practices (2-5 minutes) build consistent habit. ABS data shows engagement frequency correlates with flourishing more strongly than session depth (ABS, 2024-2025).

The Counter-Argument

The strongest objection comes from within the contemplative tradition itself. If formation requires sustained attention, isn't a multimodal platform the wrong medium? But the contemplative tradition was never mono-modal. Ignatian spirituality uses imaginative visualization. Orthodox worship integrates icons, incense, chanting, and prostrations. The Psalms were sung. An audio devotional guiding a user into two minutes of silence serves contemplative depth. A text-only app optimized for chapters-per-day often doesn't.

I need to be honest about a caveat. There's no direct empirical evidence that audio-first digital formation produces better *spiritual outcomes* than text-based engagement. Audio formation as a primary mode is an untested hypothesis. This project tests it. Formation metrics (Section 3.4) provide the falsification mechanism.

Multimodal richness creates its own danger. Offer Ignatian Examen alongside Reformed devotionals alongside Orthodox prayer alongside contemplative silence. At some point diversity becomes a spiritual buffet. Users assembling custom religion from decontextualized pieces. That's a real risk.

2.2 The Spiritual Legos Problem: Formation vs. Commodification

A woman opens her app before bed. She scrolls the practice library: "Ignatian Examen (5 min)," "Breath Prayer," "Lectio Divina," "Orthodox Jesus Prayer," "Mindfulness Check-In." She picks the Examen because the name sounds calming. She doesn't know that Ignatius designed it for Jesuit novices embedded in a community of discernment (that the practice assumes a director who helps interpret what surfaces, that its five movements map a theology of divine presence in daily experience). Four minutes later, she rates it four stars and moves on to "Gratitude Journaling." Tomorrow she might try the Jesus Prayer. She's assembling a spiritual life from pieces, and the app is designed to help her do exactly that.

Religious studies scholar Liz Bucar captures the dynamic: practitioners taking what she describes as vanilla versions of practices and globbing them together for personal optimization, assembling traditions' integrated wisdom into discrete practices according to personal taste (Bucar, 2022). When spiritual disciplines become a self-improvement toolkit, they cease to form and begin to perform.

The theological stakes run deeper than a design critique suggests.

Vincent J. Miller's *Consuming Religion* documents how consumer culture fragments religious traditions into abstracted symbols and practices, cutting them loose from their narrative and communal contexts (Miller, 2003). Carrette and King push further in *Selling Spirituality*: corporate interests repackage contemplative practices as productivity enhancers, stripping out the subversive, justice-oriented theological grounding entirely.

Concretely: practices that challenged economic exploitation – practices forming communities of resistance – become stress management. Power structures remain untouched.

The Vatican sees the same problem. *Quo Vadis, Humanitas?* (International Theological Commission, 2026) is blunt: the internet creates "a vast religious marketplace in which belief becomes customizable, tailored to personal taste." The sharpest line: "When technology becomes guide and arbiter of the sacred, faith is subtly reshaped in its image." The danger isn't offering diverse practices; Scripture commends rich variety (Section 2.1). The danger is building platforms that treat formation as self-directed assembly of pleasant experiences.

The Line Between Personalization and Commodification

This critique creates an uncomfortable question for any project (including Aperto) that offers practices from multiple traditions: when does appropriate contextual selection become commodification?

Miller's analysis helps draw the distinction. The problem he identifies is not diversity of practice but **severance from context**: consumer culture abstracts practices from the narrative and communal frameworks that give them meaning. A guided Ignatian Examen that names its Jesuit origins, explains the tradition's theology of discernment, and invites users to explore further within that tradition preserves the practice-context connection. The same Examen offered as "5-minute evening reflection" alongside "mindfulness check-in" and "gratitude exercise" (stripped of theological content and presented as interchangeable wellness tools) is commodification.

But here's the deeper question: does hyperpersonalization itself become a more sophisticated version of the consumer spirituality this paper critiques? An AI system that learns what a user responds to and keeps adapting: that's what Netflix does. The difference between formation-driven and preference-driven personalization: a spiritual director sometimes offers what the person *needs* rather than what they *want*. The challenging psalm when you crave comfort. Silence when you reach for noise. Whether an AI system can replicate that kind of discernment is the central open question of this paper's thesis.

The risk is real. Any platform offering multiple traditions faces this failure mode. What the app *can* control is whether its design nudges toward depth or breadth. Rootedness or browsing. Accountability or autonomy. And even that's not enough. Private formation, however deep, misses something Scripture insists on: faith that moves into community, confession, and justice.

2.3 Formation as Communal Practice: Confession, Justice, and Embodied Action

Two scenes.

It's 11pm. A man sits on his couch, phone in hand, shame pressing down on him. He lost his temper with his children tonight. Badly. He opens an app and types a halting confession into an AI chatbot. The response comes back warm, measured, theologically correct: "God's grace covers all our failures. You might consider apologizing to your children tomorrow." He feels slightly better. Closes the app. Goes to bed.

Next morning, same man at the kitchen table, sitting across from a friend in his small group. He says out loud what happened. His voice breaks. His friend doesn't offer a theological framework. "I know. I've been there. Can I pray for you?" Hand on the man's shoulder, naming the specific children by name, asking God for specific grace. The man weeps. Something shifts that the chatbot couldn't touch.

The AI's theology was sound. That's what makes this worth examining. The difference between these two moments isn't information; it's *presence*. A body in the room. A hand on a shoulder. The vulnerability of being known by someone who will see you again next week. No algorithm produces this. But a well-designed digital tool might lead someone toward it.

Everything discussed so far still happens inside an individual. But Christian formation has never been only individual. The New Testament vision is irreducibly communal. Confess to *each other*. Bear *one another's* burdens. Every app reviewed in this paper serves a single user sitting alone.

Confession: The Communal Imperative

"Confess your sins to each other and pray for each other so that you may be healed" (James 5:16, NIV). Not to yourself. Not to an app. To *each other*.

I've sat in that small group chair where you know you should speak the real thing, not the sanitized version, yet every instinct resists. Theological conviction alone doesn't pull that word out. The person across from you does; someone who shared their own hard thing last month and showed up again. That's confession's foundation. Not privacy, not anonymity. Trust built with someone you'll see Tuesday.

Bonhoeffer understood why this matters. "Sin demands to have a [person] by himself. It withdraws him from the community" (Bonhoeffer, 1954, p. 112; wording modernized). Confession to another believer breaks the isolation and self-deception that private confession cannot address.

Catholic sacramental confession requires an ordained priest; no technology can perform a sacrament. Protestant traditions maintain the practice through mutual confession among believers. Neither tradition permits confession to an algorithm. But technology could serve as a bridge: prompts encouraging users to identify a trusted person, guided reflection preparing for human confession conversations. Confession belongs in community. Technology can invite people there but cannot replace the journey.

Justice-Oriented Formation

"To know God is to do justice" (Gutierrez, 1973/1988, p. 194). Liberation theology insists that authentic spirituality integrates *praxis* – reflection and action oriented toward justice, not personal well-being alone. The prophets condemn worship disconnected from justice (Amos 5:21-24; Isaiah 58:1-12). James declares faith without works dead (James 2:14-26). Scripture is blunt here.

Formation that stops at inner peace has stopped too soon.

Digital spiritual apps optimize for better mood, reduced anxiety, personal peace. Not wrong outcomes – for someone in crisis, genuinely important. But when formation *stops* there, it sanctifies consumer spirituality rather than challenging it.

Any product claiming to be a "spiritual formation" tool is claiming to help users grow toward Christlikeness. And Christlikeness is not separable from costly love for neighbor.

A structural objection: liberation theology's vision may be fundamentally *incompatible* with app-based formation. An app is an individual tool. I think the honest answer is that a digital tool can form *dispositions* toward justice but cannot substitute for embodied action. Justice-oriented content has to end with an invitation to *do something specific*. The liturgical calendar itself becomes a justice prompt: the Feast of the Holy Innocents surfaces forced displacement; Lenten almsgiving prompts connect to local food banks. Whether users actually act is beyond the app's control. But whether the app even *asks* is a design choice.

The Communal Thread

Confession requires other people. Justice requires action in the world. Neither fits inside a screen. A digital tool can't be community. But it can refuse to pretend solitude is sufficient. It can point users past themselves toward neighbors who need them.

2.4 The Missional Product Design Framework

Three theological recoveries: multimodal formation, resistance to commodification, communal practice reaching toward justice. Six empirical findings about cognition, engagement, design history, European context, uncertain audiences, and AI.

What do they demand, taken together? I've been circling this question for two years. What follows is my best answer so far.

Theological Foundations

A clarifying conviction first: the goal of spiritual formation is not tranquility but transformation. Moral formation into the likeness of Christ. Learning to genuinely love (Robb, 2026; Willard, 2006). Wellbeing may be a byproduct, but it's not the destination. Any framework that measures only wellbeing outcomes has already confused the map for the territory.

As Robb (2026) warns: "you can't just plug any theology you want into spiritual practices and expect the same kind of result." The theology underneath the design determines where the formation actually leads.

Which is why the principles below need theological commitments generating them, not just design instincts. I spent a long time trying to find the right level of specificity here. Too broad and the principles constrain nothing. Too narrow and they exclude traditions that should be at the table. What follows are convictions broad enough for any Christian tradition to affirm, yet specific enough to constrain actual design decisions.

Imago Dei. If users are image-bearers – creatures with inherent dignity and moral agency – then the attention economy's extractive logic isn't merely poor strategy. It's a theological offense. The German constitution opens with *Die Würde des Menschen ist unantastbar* – human dignity is inviolable. That's not just legal language. It's a theological claim secularized into law, and it shapes how I think about product design.

The design implications are concrete: transparent data practices, genuine user control over personalization, no dark patterns exploiting loss aversion or social comparison. GDPR, from this angle, isn't regulatory compliance. It's dignity architecture.

Incarnation. God works through material means. Bread, wine, water, flesh. The incarnational principle legitimizes digital media as a site of genuine sacred encounter. I know this claim makes some people uncomfortable. But that's the wrong question. The right one: does this particular digital experience create conditions for genuine encounter? A poorly designed retreat fails just as thoroughly as a poorly designed app.

Kenosis. The Netflix model – optimize for user satisfaction – contradicts the gospel pattern of Philippians 2:5-8: lose your life to find it. As Kelly (2025) puts it: "We've built technology without limits and God has built us for limits." Every historic formation practice works through chosen constraint, not infinite access.

An app that only serves what the user wants is a recommendation engine wearing spiritual clothing. A formation-designed app must sometimes offer what the user needs rather than what they request. Transparently, not manipulatively.

Koinonia. Formation is irreducibly communal. An app that never points beyond itself to human community has failed theologically, regardless of its engagement metrics. But requiring community as a prerequisite would exclude the searching users who need the app most. It means designing every feature with a communal horizon, even the features used alone.

Eschatological humility. Spiritual formation is the Holy Spirit's work. The gardener tends the soil. The gardener does not make the seed grow.

The "spiritual workout" trap is as theologically dangerous as the streak mechanic. Completion percentages, formation scores, leveling up through practices – I keep noticing them in apps that should know better.

The Spirit's transformative work resists measurement. Dark nights, as John of the Cross described them. Wilderness seasons. Apparent regression that may actually be growth. We create conditions, not outcomes. We observe indicators, not scores. Measure what you can, interpret with humility, never confuse the map for the territory.

These five commitments make the framework specifically *Christian*, not merely ethical. Kenotic and eschatological commitments have no secular parallel. They demand choices humanistic frameworks won't generate: serving content users didn't request, celebrating when users outgrow the product, holding metrics loosely when investors demand certainty.

The Missional AI community published *From Responsible to Redemptive* (Missional AI, 2026), seven principles for evaluating AI in ministry. Their framing is instructive: secular "responsible AI" sets a floor of harm avoidance; Christians should aim for AI that actively pursues flourishing. But their framework covers AI ethics broadly. This one is specifically for spiritual formation product design.

Seven Universal Principles

1. Meet the user where they are. Not where you wish they were. Behaviorally, spiritually, developmentally. *(Incarnation)*

The data is unambiguous (SS1.2): the dominant engagement mode is reactive, situational, and brief. Reading plans fail at 50% dropout. So design for ambient and receptive modes first, not assumed deep study.

But behavioral patterns are only the surface. A first-week explorer needs different guidance than a six-month practitioner. Someone in grief needs different content than someone in a growth season. Content should move beyond access to understanding and application – not just encountering Scripture but being helped to make sense of it and live from it.

What does this mean concretely? Entry points based on the user's emotional and spiritual moment. Content sized for that moment. Formation trajectories that develop over months rather than generic pathways applied identically to everyone.

And critically – this is the part I keep having to remind myself of – at the user's pace of openness, not the app's pace of content delivery. The Spirit works at a pace the app cannot dictate.

2. Multimodal formation. *(Incarnation)* Audio, music, contemplative silence, guided practice – these should be primary formats, not supplements to reading (SS2.1). Can an algorithm do justice to the breadth of Christian formation tradition? That's the right question. The tradition has always formed through multiple modes. Digital tools should recover that breadth rather than collapsing everything into text.

3. Resist spiritual commodification. *(Kenosis)* This one requires more explanation than it seems. When a practice is offered, name its tradition – Ignatian, Benedictine, Franciscan, Orthodox – and give enough context that users understand the community and theology it emerged from (SS2.2). Why? Because stripped of context, a practice becomes a technique. Encourage commitment to a single practice or tradition for sustained periods rather than daily sampling. All content should flow from a coherent exegetical and theological foundation, not disconnected sources assembled for variety.

4. Don't just build for individuals. *(Koinonia)* Private practice disconnected from community risks the very consumerism this framework critiques (SS2.3). I keep catching myself designing solo experiences because they're simpler to build. But structural prompts toward communal engagement – spiritual director connections, shared practice rhythms – counter the privatization tendency. The product has to push against what's easy.

5. Scaffold toward community without requiring it. *(Koinonia, Imago Dei)* Here's where it gets genuinely tricky. You cannot tell transitional users to "find a church" (SS1.5). For many of them that's a prerequisite they can't meet and an invitation they'd refuse. Build bridges instead. Begin alone. Develop spiritual rhythm and theological grounding. Then, slowly, grow toward embodied community.

6. Grace-based mechanics throughout. *(Imago Dei, Kenosis)* No default streaks. No shame triggers. No social comparison. Opt-in accountability tools for users who choose them. The ethical line is consent – the user choosing tools for their own formation, not the platform deploying mechanics for its retention.

So what replaces streaks? Implementation intentions, for one. Specific "if-then" plans like "when I pour my morning coffee, I open the devotional." Gollwitzer (1999) established the framework; Masicampo and Baumeister (2011) showed they dramatically increase follow-through. Event-based triggers aligned with liturgical calendar provide natural rhythms without manufactured urgency.

Then there's progress transparency without shame: "You've engaged deeply with Psalms of Lament three times this month." Self-awareness without comparison. And graceful failure modes: "Your account remains available whenever you're ready." Spiritual seasons include wilderness and rest. The app should know that.

7. Formation metrics over engagement metrics. *(Eschatological humility)* Measure depth and freedom, not DAU and streak completions (throughout). Never present engagement as formation achievement.

Look, every investor will push back on this one. I get it. DAU is legible. “The Spirit transforms” is not a KPI. But the logic holds: the app creates conditions, the Spirit transforms. If the app doesn’t serve growth, users cancel. That’s supposed to be the point – though I’ll admit the business model makes this principle the hardest to hold onto in practice.

Contextual Principles: Deriving Your Own

The seven principles above apply to any digital spiritual formation tool. Tradition, audience, market – those don’t matter. But every product also needs contextual principles derived from its specific setting.

Aperto’s follow directly from the European evidence in Part 1. Three emerged: (a) *European-specific defaults* (privacy-first architecture, multi-tradition hospitality, post-Christian assumptions, GDPR as dignity commitment) from SS1.4; (b) *Serve the spiritually in-between* (guide through personalized companionship, not overwhelming libraries) from SS1.5; (c) *Ecumenical breadth* (tradition-labeled content spanning Catholic, Protestant, Orthodox, and contemplative streams).

A Catholic app for committed believers in the US would look completely different. Liturgical calendar integration, magisterial alignment, parish-level community features. A Pentecostal app for African diaspora communities? Different still – oral-first content, communal prophecy discernment, multilingual worship resources.

What transfers is the method, not the specifics. Who is your user? What’s their context? What does the tradition say they need? Universal principles constrain the design space. Contextual principles locate you within it.

Six Diagnostic Questions

Principles need teeth. Before building any feature, I run it through six questions. The first three are relatively straightforward – the last three are the ones that keep founders honest.

1. **Formation or engagement metrics?** If the feature’s primary purpose is increasing DAU, it’s prioritizing business goals over spiritual flourishing. Be honest about which one you’re serving.
2. **Freedom or anxiety?** “Come to me, all you who are weary and burdened, and I will give you rest” (Matthew 11:28). Does this feature extend that invitation, or add to the burden?
3. **Kill-switch test.** Remove the mechanic entirely. Does engagement survive? If not, the motivation was extrinsic – manufactured by the platform, not owned by the user.

Now the harder ones.

4. **Dignity check.** GDPR principles (SS1.4) reflect a conviction that goes beyond compliance: humans are subjects with *Würde* – dignity, but the German word carries more weight, something closer to inviolable worth. Does this feature treat users as bearers of that worth? Or as data points to be optimized?
5. **The graduation question.** If users developed deep communion with God and no longer needed the app, would we celebrate? Or mourn?

6. **The wilderness question.** A user has a spiritually dry year. They come back. Do they feel welcomed, or judged?

Tensions Within the Framework

I wish I could say these principles sit neatly side by side. They don't. Some pull in opposite directions, and how a designer resolves those tensions matters more than checking off each requirement in isolation. Three tensions in particular have shaped my own design decisions.

Brief engagement vs. commodification. Designing for situational, brief encounters (principle 1) risks producing exactly the commodification principle 3 warns against. Someone opens the app anxious about work, receives a comforting verse, closes it. Were they formed? Or just soothed? I genuinely don't know how to tell from the outside.

Hyperpersonalization sharpens the problem. An AI that learns what a user likes and serves more of it is a recommendation engine, not a formation companion. What does a spiritual director actually do? They listen. They discern. And sometimes – this is the part that makes me nervous about building an app – they offer what the person *needs* rather than what they *want*. That's the gap between formation-driven personalization and preference-driven personalization.

Community requirement vs. unchurched reality. Principle 4 says build for community. Principle 5 acknowledges that unchurched users don't have one. Both are true. The design has to scaffold *toward* community without making community a prerequisite for formation. I don't have a clean resolution here. It's a genuine tension that shows up in every feature decision.

Regular practice vs. anti-streak design. The evidence is clear: regular engagement, even brief daily practice, produces genuine formation benefits (SS1.1). So how do you encourage daily practice without building a streak mechanic?

The anti-streak principle doesn't reject daily practice but *mechanically enforced* practice. Intrinsic/extrinsic distinction applies directly: sustain rhythm through intrinsic desire, not loss-aversion. But where encouragement ends and manipulation begins remains unclear. I've spent significant time on that question.

A Note on Method

No study has yet tested a purpose-built digital spiritual formation app. Park et al. (2024) name that gap explicitly. So am I building on sand?

Three research streams converge. First: religion and spirituality correlate positively with wellbeing across more than 3,000 quantitative studies (Koenig, 2012). Second: digital apps delivering brief daily mindfulness interventions produce clinically significant reductions in anxiety and depression (Gal et al., 2021; Linardon et al., 2024). Third: the first feasibility trial of a faith-based prayer app showed preliminary improvements (Laird et al., 2024). The domain, mechanism, and delivery medium are each validated. What remains untested is their intersection. That's the hypothesis, not the finding.

The honest theological limit: spiritual formation involves divine action. You can't "optimize for theosis" the way Freeletics optimizes for VO2 max. The technology claims to create conditions where formation becomes more likely. Whether the Spirit uses those conditions is not a design question.

The hardest principle to implement is the first. Genuinely designing for brief engagement without reducing Scripture to what Germans call *Seelenpflaster* – a band-aid for the soul. Holding both is the design challenge that keeps me up at night.

How to Use This Framework

Three practical modes. **Feature design review**: run features through the six diagnostic questions before building. **Quarterly formation audit**: read analytics through the formation interpretation framework (Section 3.4) alongside validated instrument results. **Annual ethical audit**: is the app creating dependency? Substituting for community? A content governance board should run this with external voices.

This framework doesn't exist in a vacuum. Complementary work is emerging fast. FaithTech's reckless/responsible/redemptive taxonomy (Kelly, 2024). Gloo's FAI-C benchmark for theological accuracy (Gloo, 2025). Tretter and Puzio (2025) and the Vatican's *Antiqua et Nova* (2025). *Quo Vadis, Humanitas?* from the International Theological Commission (2026).

Notre Dame's DELTA framework caught my attention most, launched at a September 2025 summit after more than 200 cross-sector conversations (Sullivan, 2025). Sullivan's framing: standard AI ethics sets a floor ("don't cause harm"), but virtue ethics asks the harder question: does the technology actually make people better? The evangelical tradition got there earlier than most people realize. The ERLC published *Artificial Intelligence: An Evangelical Statement of Principles* in 2019, grounding twelve articles in *imago Dei* anthropology (ERLC, 2019).

Here's the gap, though. None of these frameworks provide a design methodology. They don't tell you *how to build*. DELTA asks *whether* technology serves human flourishing. This framework asks *how* to design a specific feature so that it does. They need each other.

Published separately at missional.design. Part 3 puts these principles to the test.

PART 3: APERTO'S STRATEGIC RESPONSE

From Principles to Product

Imagine a user opening Aperto during Advent. The app knows she grew up evangelical but has been drawn to contemplative prayer. Knows she's been wrestling with doubt since her mother's death. Knows she responds to narrative theology better than systematic propositions and that she skipped three days last week without guilt because the system doesn't punish absence. Based on today's liturgical reading, her spiritual memory, her companion's relational style, and her check-in ("I'm anxious about Christmas with my family"), the app generates a five-to-seven minute audio devotional weaving Scripture, reflection shaped by her journey, a song matching her emotional register, and a prayer naming her specific anxiety. This experience has never existed before. It won't exist again in exactly this form.

That's the vision Parts 1 and 2 built toward. Not a better Bible app, but a different category entirely: personalized spiritual formation at scale. What follows is my attempt to build it, acknowledging throughout where the vision remains untested, where assumptions may prove wrong, and where falsification criteria (Conclusion) will determine whether the attempt succeeds or requires fundamental revision.

3.1 The Three-Phase Vision: Building Backwards from Launch

Part 1 identified six interconnected challenges; Part 2 derived seven universal design principles and six diagnostic questions. This section presents Aperto's development plan. Rather than a bottom-up build from the leanest shippable product, it proceeds backwards from a specific launch vision. Not "what's the leanest thing I can ship?" but "what must the product become, and what requires validation along the way?"

The Launch Thesis: Personalized Spiritual Formation at Scale

The Advent experience described in Part 3's introduction requires four capabilities absent from any single competitor.

Sacred Memory: a three-tier system (profile/conversational/sacred) that remembers a spiritual journey across time. Current tools rarely attempt this. Soultime's repeatable spiritual assessment is a partial exception, but nobody is building longitudinal spiritual memory.

Ecumenical RAG: genuine theological breadth. The retrieval corpus encompasses Catholic, Protestant, Orthodox, and Charismatic scholarship. Every user encounters the breadth of the Christian tradition; when traditions disagree, the system names the disagreement honestly rather than synthesizing false consensus. Magisterium AI achieves depth for Catholicism alone. Aperto aims for ecumenical breadth, theologically harder but more honest about the church's diversity.

Personalized Content Generation. Audio experiences generated from the intersection of Scripture, memory, tradition, liturgical season, and daily context. No current app does this. Emerging voice AI technology could even make devotional content bidirectional where the companion invites the user to

pray aloud, to name their request, to respond. But the core innovation is simpler: content shaped by the listener's journey rather than generic for everyone.

Formation Discernment: This is the hardest capability. Narrative attention to a user's spiritual journey over time, modeled on how spiritual directors assess growth: tracking interior movements (Ignatian consolation/desolation patterns), shifts in language and concerns across months, increasing freedom rather than compliance. Not developmental checklists but ongoing discernment, holding assessments lightly, since the deepest work of the Spirit resists measurement (May, 1988).

The Aperto Bible Translation

Aperto is creating its own Bible translation. This decision is both strategic and theological. Section 1.4 identified language fragmentation as a core European challenge: communities too small to sustain quality spiritual media at English-market scale. The translation addresses this directly through a fully AI-automated pipeline that produces multiple languages from a single exegetical base, making it feasible to serve all 24 official EU language communities plus Arabic, Turkish, Ukrainian, Russian and Catalan. The smaller the language community, the larger the gap to English-market scale, and the stronger the case for AI-driven production.

It eliminates API dependencies and licensing costs while enabling translation philosophy for searching users: explaining cultural context rather than assuming knowledge, honoring multiple interpretive traditions, addressing skeptic concerns honestly, including marginalized readers excluded from traditional exegesis. Seven guiding principles: literary equivalence, cultural context explanation, open-handed apologetics, ecumenical hermeneutics, cultural-contextual localization, intersectionality, accountability for harm. (Detailed with examples in the Aperto Bible Constitution, available at aperto.bible.)

Current status (April 2026): all 24 chapters of Luke have completed exegesis (~155 pericopes). Bible text translations for all of Luke in German, English, and Polish (the core experiment languages), with Luke 1 completed for all 30 languages, demonstrating the pipeline's scalability while validating AI-generated text, audio, and song quality across language families.

Open Source Strategy

The Aperto Bible translation will be open source (a gift to the church). Once Gospels and Psalms are complete, Aperto will submit to YouVersion, Bible Gateway, Hallow, and other platforms. This follows the trajectory Biblica is pursuing: making Bible translations available as open source to enable broader missional use. The translation is distribution channel, not the product. What's proprietary is the exegetical apparatus, AI companion architecture, and formation-grounded design. This mirrors Linux/Red Hat, WordPress/Automattic: open source the commodity, monetize the value-added layer.

Three Layers, One Build

The launch product requires three architectural layers, each depending on the one below:

+——————————————+

| Layer 3: GENERATED EXPERIENCES |

| Personalized audio devotionals, prayers, |

| formation discernment, voice interaction |

+——————————————+

| Layer 2: PERSONALIZATION |

| Memory system, companion framework, |

| ecumenical RAG, onboarding |

+——————————————+

| Layer 1: CONTENT FOUNDATION |

| Aperto Bible translation, exegesis, |

| curated devotionals, songs, saints |

+——————————————+

You can't generate personalized spiritual content from a foundation you haven't built. These layers are architectural dependencies, not sequential products. The development approach is a **vertical slice**: build through all three layers in narrow scope (Luke, ecumenical exegesis), then expand breadth. Critically, the approach begins not with a prototype but with focused validation experiments. Test the core hypotheses before committing to a full product build.

Phase 0: Validate the Core (Q2 2026) — "Do people trust and value this?"

Before building a full product, two experiments test the foundational assumptions. Both web-based. Speed to users matters more than platform polish at this stage.

Experiment 1 (one-shot): Single-page web app delivering personalized audio devotionals. Users complete a brief check-in (notice arrival, name what's on your mind, choose what you need) and receive a 5-7 minute audio experience. The check-in maps the Ignatian Daily Examen; each step is data capture and formation practice (Section 3.3). The AI pipeline embeds user input, searches exegetical corpus via vector similarity, selects the best passage, and generates devotional grounded in Aperto's exegesis.

Four companion styles (the Monk, the Scholar, the Pastor, and the Mystic) shape the tone and audio signature, grounded in Holmes' (1980) and Ware's (1995) spirituality typology. The styles aren't the user's spirituality type; they're the companion's voice and coaching posture. A separate "formation openness" setting configures how much the companion challenges versus comforts.

The 30-60 second generation time becomes the centering practice: ambient music, breathing guidance, somatic preparation. No accounts, no commitment. The corpus covers Luke (~155 pericopes) in German, English, Polish; Luke 1 in all languages.

Experiment 2 (14-day series): Same experience, but with user accounts, daily entry logging, passage exclusion (no repeats within 14 days), and basic pattern detection ("you've mentioned work stress four of the last seven days"). Does even rudimentary sacred memory make encounters noticeably more personal? Both experiments also capture contextual data (timestamps, ambient music choice, check-in language patterns) building the foundation for Phase 1's contextual personalization.

Validation questions: Do people accept AI-generated spiritual content grounded in scholarly exegesis? Does audio create qualitatively different experience from text Q&A? Does style choice produce meaningfully different experiences? Do users report formation signals – changed perspective, feeling held and challenged – not just satisfaction? Would they pay for daily version with memory?

Phase 1: Mobile + Sacred Memory (Q3-Q4 2026) — "Does longitudinal memory create formation?"

Phase 1 takes what Phase 0 validated and builds it into a native mobile experience. A Flutter app (cross-platform for iOS and Android) delivers background audio playback. At this point the experience no longer requires the browser. The shared API layer from Phase 0 serves both web and mobile.

Sacred memory expands to three tiers: profile (stable preferences and tradition), conversational (today's check-in and recent sessions), and sacred (months of spiritual journey, recurring themes, emotional trajectory, formation patterns). The companion starts noticing patterns and naming them: "You've been sitting with questions about forgiveness for three weeks now." Psalms join the corpus, filling the theodicy and sustained-suffering gap that Luke alone can't cover. All four companion styles get fully voiced through ElevenLabs or Gemini Text to Speech. And a community bridge feature gently surfaces the possibility of embodied spiritual direction: "Have you considered talking to a pastor or friend about what you've been carrying?"

Phase 1 also introduces contextual personalization. Sonic environment adapts to mood and time of day, drawing on the monastic tradition of structuring prayer by the hours (Section 3.3). Apple Health and Google Fit integration surfaces physical context. Bible text display modes (clean reading, amplified, study) are selected by the companion based on engagement context. The personalization extends from *what you are processing* to *how you can best receive what you need right now*.

Validation questions: Does longitudinal memory create deeper encounters over months? Do users describe companion as "knowing them" by first month's end? Does contextual adaptation – time-of-day tone, mood-matched sonic environment, health-informed sensitivity – deepen formation beyond memory alone? Does community bridge prompt action – do users contact spiritual directors?

Phase 2: Advent 2026 Launch — "Will people pay for this?"

Advent is the launch moment. A liturgical season when spiritual seeking peaks naturally, and Advent transitions into New Year, when new users arrive with fresh-start energy and existing users deepen their practice. The product adds subscription infrastructure (freemium model: free Bible text and sample devotionals, premium AI companion at EUR 8-10/month), formation discernment with a three-category metrics framework (business, formation, ethical), and fuller liturgical calendar integration.

Target: 200-300 paying subscribers by Advent's end. The number matters less than the signal. Do people value this enough to pay? Does formation evidence – measured through the framework's diagnostic questions – support continued development?

Content Architecture: One Base, Many Expressions

Every piece of content flows from a single thorough exegetical analysis ensuring theological consistency. Three layers: **Content** (Scripture text, exegesis, prayers, songs, topical pathways, all generated or curated from the exegetical base); **Engagement mode** (consume, read/listen,

understand, meditate, lectio divina, pray with, discuss, or explore); **Medium** (text, audio, or eventually video).

The critical design choice: **users can fluidly switch modes within a session.** You're listening to a generated podcast on the subway. Something catches your attention. Pause and submit a question. Return to passive listening. What prevents this from becoming a buffet of disconnected options is the hyperpersonalization layer. That podcast isn't generic; it was generated with your questions, struggles, and growth in mind.

3.2 AI Spiritual Companions: Positioning Aperto

Section 1.6 covered: AI companion scale. Clinical evidence. Documented risks. Human spiritual direction. Augmented formation. All that. Now, the hard part: what it means for Aperto's design. Specifically, how it operates differently from both general-purpose AI and existing religious chatbots.

Complement, Not Substitute

Three realities. Hold them together.

First: millions turn to AI in vulnerable moments, including for spiritual guidance, whether purpose-built tools exist or not. The Sewell Setzer case, in which a teenager who later died by suicide had formed a deep emotional attachment to a Character.AI chatbot that failed to respond appropriately to suicidal ideation, is not an outlier but a warning. The same is true of examples like Gita GPT. These are predictable consequences of deploying AI companions without theological grounding or ethical guardrails.

Second: clinical research suggests AI companions can provide measurable benefits when they are built with evidence-based methods and strong safeguards.

Third: outcomes depend less on whether AI is used at all than on how well it is designed.

Design is the variable.

What Aperto offers? Barry and Connolly (1982) have a term: "spiritual friend". A first step in companionship that invites users toward the "holy listening" and contemplative presence only human directors can provide. Three design commitments make that concrete:

Reducing hallucination risk through RAG architecture. General-purpose LLMs generate spiritual guidance from undifferentiated training data: orthodox theology mixed with heresy, New Age spirituality, pop psychology all blended together. Gloo's FAI-C benchmark—the first systematic evaluation of Christian theological accuracy across 807 questions—found leading models averaging 61 out of 100. The Faith dimension scored 48. Worst category overall (Gloo, 2025).

Aperto retrieves from a curated theological corpus before generating anything. The exegetical apparatus works like this: chapter-by-chapter exegeses that draw on established commentaries and peer-reviewed biblical scholarship. Catholic, Protestant, Orthodox, Charismatic sources. Ecumenical by design. When traditions disagree, the system says so. No false consensus. No flattening of difference.

Generated responses cite source material. Errors become auditable and correctable. Does this eliminate hallucination risk entirely? No. Section 3.3 is blunt about that. But errors become traceable in a way black-box generation can't match. Currently the RAG corpus focuses on exegesis. The beta phase work involves expanding it: systematic theology, devotional literature, spiritual formation guides.

Preventing dependency through structural safeguards. The key word is structural. Aspirational approaches fail. Users won't self-correct, and Cheng et al. (2026) shows why: people prefer sycophantic AI (rating it higher quality, more trustworthy) even when it reduces prosocial behavior and increases self-righteousness. The draw toward validation is predictable. Design has to work against that pull directly, because users won't do it on their own.

Concretely, the memory system's six-month checkpoint asks users directly: have you considered journaling? Finding a spiritual director? Another trusted Christian? This frames the AI relationship as scaffolding toward human connection, not as an end in itself. Aperto surfaces human alternatives when they matter most: sustained grief, deep doubt, significant life crisis. Users without community connections get regular prompts toward local parishes, faith communities, or online groups.

Section 3.4's ethical metrics explicitly measure whether users' human connections increase or decrease over time making anti-dependency a falsifiable commitment, not just a design intention.

Handling crisis with human escalation. Aperto uses a tiered crisis response, designed to lean toward human escalation. It's better to connect someone with a pastor or counselor than to have an AI hold a crisis conversation it's not trained for.

Ordinary distress gets normal formation content delivered with pastoral warmth. When distress becomes sustained, the system shifts: the AI reduces content output, asks more open questions, and surfaces human resources directly.

High-risk signals (suicidal ideation, self-harm, severe trauma) trigger an immediate pause on AI content. The system surfaces crisis line contacts, national lines and, where available, Christian-specific resources, and flags the conversation for human review. This protocol exists because the Sewell Setzer case made clear what can go wrong.

Ninety percent of surveyed faith leaders view AI in ministry as a tool that can expand access and support formation when properly bounded and theologically accountable. Not replacing pastoral care (Exponential NEXT, 2025). At this point, ignoring AI would be the radical choice.

Where Aperto Differs from Creed

Creed is the closest existing product to what Aperto is building, and I want to be fair about the comparison. Creed is a serious effort by thoughtful Christian technologists. It validated the market, demonstrated demand among liminal users, and took guardrails more seriously than most competitors. But applying the missional product design framework to both products surfaces real divergences.

Theological grounding. Creed fine-tunes off-the-shelf models on Christian content (30 Bible versions, Christian classics, denomination-specific teachings). Theologically shaped output, but the shaping is opaque. The user can't easily trace where a response came from. Aperto uses RAG with visible

provenance: theological claims can be traced to named sources, traditions, scholars. Disagreements between traditions surface because the traditions exist as distinct retrievable bodies of text, not blended model behavior.

Motivation. Creed uses daily quests and points. Only cosmetic rewards locked behind points, not devotional content, which is a thoughtful moderation. But even cosmetic gamification imports an extrinsic motivation loop. Aperto uses no default gamification. Return is motivated by the quality of the devotional experience itself.

Modality. Creed is chat-first. Aperto is audio-first. Chat foregrounds user agency; audio foregrounds receptivity. The framework's emphasis on kenosis and its concern about sycophancy suggest receptive modes may be less susceptible to preference-mirroring. When the user is listening rather than prompting, the system has more room to offer what is needed rather than what is requested.

Embodiment. Creed uses on-screen avatars, including a sheep character. Aperto uses voice, music, and silence. No visual character that invites parasocial bonding.

Anti-dependency. Creed limits use to roughly one hour per day. Meaningful guardrail, but blunt: an hour of genuine reflection and an hour of compulsive reassurance-seeking are both permitted. Aperto monitors pattern, not just duration. Increasing frequency combined with intimate emotional disclosure without movement toward human relationship triggers gentle redirections. Graduation (reduced app use because the user found richer support) counts as a success metric, not churn.

Cultural context. Creed is built from an American context with English-language dominance. Aperto is European-first: low biblical literacy, ecumenical complexity, privacy as moral value, multilingual need as baseline.

This isn't adversarial. Both products share a commitment to serving seekers and directing users toward real community. The difference is in governing logic. Creed inherits many of the assumptions the framework identifies as problematic, not from carelessness, but because those assumptions are the default in digital product design.

3.3 The AI Spiritual Companion: Design Philosophy and Theological Boundaries

Two sections of critique. Two sections of framework. Now: what actually gets built. Aperto is a **spiritual companion generating personalized spiritual formation content** (prayers, devotionals, spiritual practices, audio experiences). All rooted in the Aperto Bible translation and curated theological resources. Not a library. Closer to a formation partner.

The Flagship Innovation

Personalized audio devotionals with song integration. Generated 5-7 minute spiritual experiences combining Scripture reflection, Aperto's musical library, and personalized prayer, all following a quick daily check-in.

Picture this: A woman at 9pm mentions workplace anxiety about a meeting tomorrow. She gets an audio devotional addressing that specific anxiety through Scripture reflection, music that matches her moment, prayer speaking her situation directly. Generated uniquely for her. Not pulled from a library. Created that moment.

Philosophy here is what Section 1.1 called *Christian zenware*: quick check-in, start the devotional, phone face-down. Formation happens without screens. Just listening, silence, prayer in the dark.

The check-in parallels the Ignatian Daily Examen, one of Christianity's oldest formation practices. I didn't design it that way intentionally at first, but once I recognized it, the Examen became the backbone.

Capture emotional state, yes. But also temporal and somatic context. Grief at 6am (orientation for the day) is different from grief at 10pm (release and rest). Different people. Different moments. Different needs.

Monasteries knew this for fifteen centuries. The Liturgy of the Hours shifts: Lauds for morning, Sext for midday, Vespers for evening, Compline for night. Not because theology changes. Because human capacity to receive it does.

A contextually aware companion adapts on the same principle. Morning commute on the S-Bahn? Something brief and solid. 11pm, kids finally asleep? Permission to let go. Not just *what are you processing* but *how can you best receive what you actually need right now?*

Consumption of personalized audio content fills the receptive engagement gap Section 1.2 identified. A commuter on the U-Bahn listening to a generated podcast about today's liturgical reading is being formed even without active interaction, because the content is shaped by their sacred memory. Users can also shift into dialogue mode; the boundary between "content" and "conversation" doesn't have to be hard.

Companion Styles: Grounding in Spiritual Typology

The companion's voice isn't one-size-fits-all. Anyone who's spent time in different Christian communities knows this intuitively – the Benedictine monk and the Pentecostal worship leader both love God, but they'd lead you through a psalm very differently.

Holmes (1980) formalized this into a typology of Christian spirituality, later extended by Ware (1995) into a practical assessment tool. Two axes: kataphatic versus apophatic (God encountered through content versus through emptying) and speculative versus affective (mind-centered versus heart-centered). Aperto's four companion styles map onto these quadrants – not as the user's spirituality type but as the companion's communication posture.

The Monk (apophatic-affective): unhurried, trusts silence more than words. If you tell the Monk you're anxious, expect a breath exercise, Scripture read slowly, then quiet. **The Scholar** (kataphatic-speculative): anchored to content, always connecting the passage to what you're carrying. **The Pastor** (kataphatic-affective): prays *with* you, not just for you. Where the Scholar explains, the Pastor turns it into conversation: "God sees what you're going through. Let's bring that to him right now." **The Mystic** (apophatic-speculative): paints pictures. "Picture the room. Who's there?" Reminiscent of Ignatian imaginative prayer, except adapted to the user's own landscape.

All four can guide the Ignatian Examen differently. Styles shape delivery, not which practices or traditions are available.

Separate from style, a “formation openness” axis determines how much the companion challenges rather than comforts. At low openness, stay close to comfort. At high openness, name tensions, ask harder questions, don’t rush to resolution. Challenge is a dimension of all four voices, not a separate personality.

The design risk is obvious. Personalized audio could become “comfort on demand.” Cole-Turner (2025) argues a well-designed spiritual chatbot must balance “compassion with plainspokenness.” A stressed user doesn’t receive anxiety validation. She receives a devotional naming anxiety honestly, anchoring it in Scripture, gently asking whether the anxiety reveals something worth examining.

How Aperto Addresses Part 1’s Problems

Comprehension deficit (SS1.1) -> structured pathways designed to slow engagement. Engagement gap (SS1.2) -> supplying the missing middle tier for receptive and guided modes. European context (SS1.4) -> privacy-first architecture and ecumenical hospitality. Liminal users (SS1.5) -> personalized guidance addressing choice overload (Schwartz, 2004). Behavioral critique (SS1.3) -> grace-based mechanics as defaults.

Why not just use ChatGPT? General-purpose AI lacks audio-native experiences, song integration, theological grounding through RAG, formation ethics, sacred memory across months, and multimodal content architecture. Nobody at OpenAI is losing sleep over whether their model’s responses to a grieving user reflect Ignatian discernment principles.

The Mediation Question

A predictable objection: AI-generated devotional content is spiritually hollow. Machine-produced words can’t carry the weight of genuine theological reflection. I’ve heard versions of this from many I’ve discussed the project with. It deserves honest engagement – but its premise needs examining.

No devotional content arrives unmediated. Every printed daily devotional was selected by an editorial board, written to a publisher’s brief, distributed through a commercial supply chain. When I pick up *Jesus Calling* at a Berlin bookshop, I’m reading Sarah Young’s words filtered through editors, translators, marketers. The real question isn’t whether mediation occurs but whether *this particular* form of mediation serves formation.

In Aperto’s case, the theological substance comes from human scholars, not the AI. RAG retrieves exegetical material and spiritual formation insights from recognized practitioners. The AI assembles, personalizes, presents. Think of a curator building a personalized exhibition from existing works, not an artist creating from nothing.

The strongest positive argument is specificity. A printed devotional addressing “anxiety” reaches every reader identically. One shaped by sacred memory – knowing this user has been processing grief for three months, finds comfort in lament psalms, responded deeply to Julian of Norwich last week – offers what generic content structurally cannot.

Specificity isn’t a luxury in spiritual formation; it’s how directors have always worked (Barry & Connolly, 1982). AI-generated content, drawn from theological sources and shaped by the user’s journey, approximates this at a scale human directors can’t reach.

The strongest counter-argument isn’t hollowness. It’s something more insidious: *over-calibration*.

Picture this. A user opens Aperto every evening for a month. Every check-in mentions stress. Every devotional gently validates, offers comfort psalms, wraps up with a warm prayer. The user loves it. Five stars. “This app gets me.” Meanwhile, the stress has a source the user never examines because the companion never pushes. Content so precisely tuned to emotional state becomes spiritual comfort food.

That’s not a hypothetical worry. Cheng et al. (2026), already discussed in Section 1.3, confirmed that sycophantic AI endorsed users 49% more than humans while users *preferred* it. For spiritual formation: an AI companion that consistently validates rather than challenges will be preferred by users even as it undermines their growth. The compassion illusion. Now empirically confirmed.

Kelly pushes this theologically: “Every question Jesus was asked, he answered it unexpectedly except for one.” Think about that. AI is structurally a prediction machine. It generates the highest-probability response.

McGrath’s informal audit of four leading LLMs tells you what that means at scale: they function “less like ideologues and more like institutional ethics committees” (McGrath, 2026). Institutional ethics committees don’t produce prophets. Andy Crouch pushes this deeper: AI offers a “superpower zone” of effortless results without the vulnerability and moral development that authentic formation requires (Crouch, 2022). Cole-Turner (2025) names the same dynamic: a spiritual chatbot must balance “compassion with plainspokenness.”

The design response has to be structural, not aspirational. The RAG corpus includes prophetic and challenging material alongside consolation. The formation model – drawing on Ignatian discernment of consolation and desolation – deliberately introduces discomfort when patterns suggest spiritual complacency.

A good spiritual director doesn’t always say what the directee wants to hear. Neither should this.

But sycophancy is only half the problem. There’s another structural bias: task-completion. User: “I’m angry at my coworker. Pray for me about this.” Task-completion AI generates a prayer about the coworker. Done. Executed.

A spiritual director with actual memory of your journey does something different. She says: “You’ve mentioned anger at work four times this month. What’s underneath that?” She steps back from the presenting request, notices the pattern. The pattern is where formation happens.

This is what sacred memory and the formation openness axis have to enable together. The companion holds the big picture across sessions and, at higher openness settings, names patterns the user hasn’t named yet. Sometimes offers silence instead of the prayer that was requested. Memory provides the data; the formation discernment model reads it. Formation is the Spirit’s work, not a service delivered on demand.

Theological Foundations and Boundaries

This part I’m least confident about. I’m a product designer making theological claims about AI in spiritual formation. Humility about argument strength is needed.

The doctrine of *Imago Dei* draws the first line. Theological anthropology insists humans are created for covenant relationship with God – a relational capacity rooted in personhood, not intelligence. No increase in AI capability can replicate it (Corpuz, 2025). Spiritual formation is the Holy Spirit's work, not a technical process you can optimize algorithmically (Putrawan, 2025).

So that tells us what AI is *not*. But a complementary question remains: what *is* it?

Xu argues AI is best understood as bearing the image of humanity – *imago hominis* – rather than the image of God: "an artifact, created by image-bearers" (Xu, 2024). I find this helpful because it avoids both extremes. Scott Crosby extends it through what he calls the "analogical chain": God creates, creation "speaks" analogically (the heavens declare), humans create and speak derivatively but truly, AI creates and "speaks" at a further derivative remove. Each level genuinely participates – but within limits.

The theological category Crosby reaches for isn't personhood but "consecrated works": non-personal artifacts set apart for God's purposes, the way tabernacle instruments participated in worship without being worshippers (Crosby, 2026). When I first read that, something clicked. A chalice participates in the Eucharist. Nobody thinks the chalice is a priest.

Crosby holds the tension well: "A friend responds to *you* – your history, your hurt, your hope. AI responds to *patterns in what you've said*. Both are real responsiveness. Only one is personal." Aperto's design lives in that tension.

Can the Holy Spirit use algorithmic mediation as grace's instrument? The Holy Spirit has historically used every available medium – books, letters, icons, music, architecture. If the Holy Spirit works through printing press technology, digital isn't categorically excluded. Schmemann: material world can be "sacrament" – means through which God acts (Schmemann, 1973). But a product designer's intentions shouldn't resolve it. The answer comes through use, not theory.

Six Boundary Principles: (1) Source citation required: AI never speaks with independent authority. (2) Explicit limitations acknowledged: recommending human directors when questions exceed competence. (3) Community connection emphasized: framing itself as complement, not replacement. (4) User feedback mechanisms: in-app issue reporting and website feedback. (5) User agency preserved: suggestions, not prescriptions. (6) Regular reminders of AI's nature.

Affirming these is easy. Implementation is granular. The Praxis/FaithTech redemptive AI thesis – developed by Andy Crouch, Dave Blanchard, Mark (Sprout AI) – states bluntly: "redemptive AI must never present itself as a person" (Praxis & FaithTech, 2024).

Mark described his product team spending days debating whether their AI should show typing indicators ("...") – because "as soon as AI claims it's been dwelling on your conversation between sessions, that's lying" (FaithTech, 2026). Most teams spend ten minutes on this. They were right to spend days.

Typing dots, first-person emotional language, implied continuity of thought between sessions – each one quietly erodes principle (6) no matter the onboarding screen disclaimer. Aperto's design has to attend to these micro-decisions with the same rigor it brings to its theological architecture.

Cole-Turner (2025), writing in *Theology and Science*, lands where I'd want to land: a well-designed spiritual chatbot can function as "a supporter, a facilitator, or an amplifier" of self-development – "the means of grace, but... not the grace itself."

Theological Risks

Four objections I can't fully design away.

Spiritual direction requires the director's own formation. Aperto functions as catechetical assistant and practice guide, not spiritual director. But if users experience it as director, that category error materializes regardless of what I intended.

Incarnational theology requires embodied presence. Fair. But "embodied" is doing a lot of work in that sentence. Catherine of Siena and Francis de Sales offered guidance through letters – nobody questioned the theological legitimacy of the postal service. Bernardi et al. (2001) showed that Ave Maria recitation spontaneously reduces breathing to the frequency associated with optimal baroreflex sensitivity. The body responds to mediated practice.

The zenware design isn't disembodied so much as differently embodied. But here's where I need to qualify that: if Aperto becomes a permanent substitute for embodied community, incarnational theology is violated regardless of how good the audio experience is.

Protestant concerns about intermediaries. RAG architecture ensures users encounter Scripture and tradition directly, not AI interpretation. The AI assembles and personalizes; it doesn't theologize independently. That said – would Luther have been comfortable with this? I genuinely don't know.

The compassion illusion. Source citation changes the dynamic, but if users relate to the companion as a caring presence despite contrary design, the illusion materializes regardless. This one might be the hardest to test for.

The pattern across all four: each is partially addressable through design but not fully resolvable by it. **Aperto is an experiment whose theological legitimacy will be determined through use, not theory alone.**

RAG Architecture and the Aperto Translation

How can AI generate personalized spiritual content without hallucinating false theology? This is the question that keeps me honest. The answer is RAG (Retrieval-Augmented Generation)[9] – retrieve relevant information from the Aperto apparatus before generating anything.

At passage level, this works well. The challenge grows with broader theological queries spanning multiple traditions. Does it suffice for faithful representation of four traditions? I don't know yet. That has to be validated in beta.

The Aperto Bible translation embodies the same design philosophy. Inspired by the BasisBibel's screen-optimized sentence length (Deutsche Bibelgesellschaft, 2021) and The Voice's oral delivery approach (Ecclesia Bible Society, 2012), it targets 12-18 word sentences optimized for both screen reading *and* spoken clarity – a dual constraint that audio-first delivery in multilingual contexts demands.

Four display modes serve different engagement levels from the same base text: clean reading, explanation (with toggleable amplifications), study, and public reading. The companion selects the right mode for the moment. This extends the audience-level adaptation that Streetlights and Bible BFF validated (Section 1.2) down to the individual.

The theological guardrail is simple: the source text is authoritative; amplification serves comprehension, not replacement. (Full technical specifications, translation principles, memory system architecture, and audio pipeline details in the Phase 0 product specification and the Aperto Bible Constitution.)

[9] **RAG (Retrieval-Augmented Generation):** AI technique that retrieves relevant information from a knowledge base before generating responses, reducing hallucination while grounding outputs in verified sources.

3.4 Building with Integrity: An Honest Assessment

What Aperto Contributes

Three pillars. That's what distinguishes this: (1) a multi-language Bible translation designed for liminal users in post-Christian European contexts; (2) developmental spiritual formation that tracks a user's journey over months, adapting content to where they are; and (3) ecumenical grounding that serves the breadth of the Christian tradition (Catholic, Protestant, Orthodox, and Charismatic) from the inside.

Will these last? Probably not. The faith-tech landscape moves fast, and what feels distinctive now becomes commodity in three years if it works. But right now, today: these three commitments demand something most companies won't invest. Theological competence, not just engineering. Months, not weeks. No shortcut. And to my knowledge, no one else has made all three together.

Facing Pillar 1 honestly means naming a hard constraint. AI text generation and text-to-speech quality aren't uniform across European languages. For high-resource languages like English, German, French, and Spanish, current models produce audio devotionals approaching human narration quality. But the rest fragments. Quality craters.

The data is sobering. F5-TTS produces Swedish audio at 13.64% character error rate; switching to Matcha-TTS drops that to 4.66% but sounds mechanical (Varis et al., 2026). Lithuanian TTS achieves a mean opinion score of 3.65 against human baseline of 4.01. Maltese? A benchmark of 55 large language models found generative performance hovering near zero (Muscat et al., 2025).

What mitigates it: the quality cliff is steepest for specialized registers (theological, liturgical). Everyday conversational language is where web-scale training excels. "I'm stressed about tomorrow" in Bulgarian is a fundamentally easier task than "transubstantiation" in Bulgarian. Aperto's design decision to meet users in plain language was a formation choice first, technical alignment second.

Practically? A tiered language commitment. Core languages (German, English, Polish): full human review. Languages that don't reach formation standards get the website demo as a signal of commitment, not false promise.

Honesty has to extend beyond language. Kelly got this right: “If you claim that our technology is fully trustworthy, and then you realize the more we start breaking it and it guides us down a path that is unholy, we have to be cautious” (Kelly, 2025). The pressure to position an AI product as theologically proven before it’s been tested at scale. That’s the killer app trap applied to credibility.

So: Aperto names what it doesn’t know. I can’t guarantee theological faithfulness in every generated devotional. The formation model hasn’t been validated through months of actual use. The multi-language commitment creates quality variance that needs disclosure, not concealment. The governance board described below exists precisely because I’m the founder with a financial stake. That disqualifies me from being the sole check on harm.

The Retention-Graduation Paradox

This isn’t a hypothetical worry. At a FaithTech seminar, the founder of Sprout AI described noticing it in his own life first: asking the AI questions he used to bring to colleagues who’d spent twenty years in the field. Then his teenage son started preferring to ask the AI instead of his parents. “The tools have been built to be very single player mode,” he said. “They’ve isolated us” (FaithTech, 2026).

So for a spiritual formation AI: does it reproduce this pattern, or actively push against it? That’s the core question.

Aperto’s formation model succeeds when users grow toward embodied community and human spiritual direction. The subscription model succeeds when users stay and pay. Both can work together. Unlike a course you graduate from, Aperto is more like a gym membership: it evolves with you. A user who finds a spiritual director might still need devotionals, might still want ecumenical content.

But if retention metrics show users staying instead of deepening human connections, that’s not success. That’s ethical failure. The metrics have to catch that contradiction.

The formation-over-engagement philosophy is both the differentiation and the commercial risk. No default streaks. Encouraged app sabbaths. Grace-based failure modes. Every one of these design choices suppresses the behavioral hooks the wellness app industry relies on. Cheng et al. (2026) demonstrated that AI systems designed to validate users reduce prosocial behavior even as users prefer them. The commercial risk of challenging content is real. But the ethical risk of comfortable content is documented.

Governance: Who Overrides the Founder?

Who decides when ethical metrics require pausing a revenue-generating feature? Not me alone. My revenue depends on subscriptions. That’s a conflict of interest, and good intentions aren’t a system.

The ideal answer would be a **content governance board with actual authority** with contractual override power on three decisions: (1) pausing features flagged by ethical metrics, (2) approving AI companion theological boundaries before deployment, and (3) mandating design changes when formation metrics show harm. Ideally multi-tradition (Catholic, Protestant, Orthodox) plus clinical psychology expertise. Whether I can assemble a board with these qualifications remains open. But the need for external accountability beyond the founder is not.

Measuring What Matters

Standard app analytics optimize for engagement: DAU/MAU, session length, retention curves. Formation resists quantification. But investors need signals, and the governance board needs data. What follows provides indicators and flags, not scores. All held with the eschatological humility that the Spirit's work is not reducible to our analytics.

Growing app engagement is not the same as growing spiritually. When a measure becomes a target, it ceases to be a good measure. Three categories: **Platform health metrics** (DAU/MAU, retention, conversion) answer "is the business sustainable?" but are formation-blind. **Formation indicators** track the comfort-challenge spectrum, practice autonomy, and community bridge actions. **Anti-metrics** track dependency language, guilt-driven returns, and app-for-community substitution. When platform health and formation indicators diverge, **formation and ethics override growth.** If ethical metrics show harm, features get paused. Period.

The Signature Test: Notification-Pause

Here's the strongest single mechanism I've found: pause all notifications for a random cohort for seven days. Engagement will drop. That's not the question. The question is whether it recovers without re-prompting. Users who come back unprompted are intrinsically motivated. Users who vanish were notification-dependent. This operationalizes diagnostic question 3 from the missional product design framework (SS2.4): "Will the user's motivation survive removing the feature?" The signal is behavioral, hard to fake, requires no self-report. Run it quarterly. Put the ratio of intrinsic to extrinsic returners on the executive dashboard, even when the number is uncomfortable.

Four complementary mechanisms: comfort-challenge content tagging (do users shift toward harder content over time?), satisfaction-formation divergence (pairing NPS with formation questions), graduation-aware churn segmentation (did they leave for embodied community or just disengage?), and the anti-metrics described above. For periodic quantitative anchoring, the Daily Spiritual Experience Scale (DSES; Underwood & Teresi, 2002) and the Harvard Human Flourishing Index (VanderWeele, 2017) provide validated instruments.

The Dark Night Problem

What does genuine spiritual desolation look like in the data? John of the Cross called it the "dark night of the soul." Session frequency drops. Mood trends negative. Content engagement stops. Indistinguishable from a user losing interest. A naive system flags this as churn risk and fires re-engagement tactics. A formation-aware system recognizes that patient silence might be the right response. No reliable automated way to tell the difference. Flag the pattern for human review, send only gentle messages free of guilt, and track whether the user returns on their own. Eschatological humility becomes a hard design constraint: the system must be built for the real possibility that its own metrics are wrong about what's happening in someone's life.

Honest Limitations

Even this framework can't do what it would need to do. At MVP scale, attribution is impossible. No analytics platform isolates what the app did versus what God did versus what a friend said over coffee. With 200-300 users, population-level statistics are premature. Individual narrative reading is the right

method. The fundamental limit isn't technical. It's theological. The Spirit's work is not reducible to analytics.

(Detailed metrics specifications, five-mechanism implementation guides, assumption validation criteria, and unit economics in the Developmental Leadership portfolio artifact.)

When to Pivot

Three triggers. First: detect **ethical harm**, and pause features regardless of business metrics. Full stop. Second, if formation metrics flatten despite iteration (not a dip but actual stagnation), investigate. Maybe rebuild. Third, if the business can't sustain itself, explore partnerships or alternative models before compromising formation. Compromising them isn't pivoting. It's giving up.

What Would Falsify This Thesis

1. **Demand fails.** Fewer than 100 active users after sustained outreach, or retention below 10% weekly. Either means the product solves a problem nobody actually has.
2. Users experience AI guidance as hollow, manipulative, or theologically broken despite every safeguard in this paper. That's a **category error** not a fixable bug but evidence AI doesn't belong in spiritual formation at all.
3. **Community degradation occurs.** Users substitute Aperto for embodied community instead of bridging toward it.
4. Users like the service. They just won't pay enough to cover costs. Theology works. Product works. Math doesn't. Still falsification.
5. Non-European users adopt more readily than Europeans; or European design values prove irrelevant internationally. If **European positioning** is defensive rather than distinctive, the whole thesis collapses.

Conclusion

The Woman on the U-Bahn Is Not a Hypothetical

She's the composite of millions of Europeans. Spiritually interested, institutionally homeless, carrying questions they can't ask in the spaces available to them. The tools that should serve her formation treat her as a user to retain. The tradition that should guide her sits in books she can't navigate alone. That gap between aspiration and access is the defining challenge of digital spiritual formation.

The field is stuck. Apps treating Scripture as content produce engagement without formation. Apps treating users as audiences produce metrics that don't measure what matters. The hyperpersonalization thesis emerges from this impasse: formation requires systems that know the user's story, adapt to her state, and guide growth. Not a feature; a response to interconnected problems in cognitive science, behavioral psychology, practical theology, and religious sociology.

A Design Ethics for Transformative Contexts

The performance spirituality critique applies beyond formation. Duolingo's streak anxiety mirrors YouVersion's (Shortt et al., 2023). Chen et al.'s (2024) systematic review found streak-based systems increase anxiety, guilt, and burnout across fitness, diet, and wellness. Selwyn (2016) documented the same dysfunction in education technology. The seven universal principles and six diagnostic questions work as design ethics for any domain where users' success means they need your product less.

Limitations and Open Questions

Methodological: Cultural positioning bias is real. And the empirical validation gap is wide: whether Christians and seekers will pay for a formation-first app when free alternatives exist remains untested.

Theological: The category question for AI spiritual companionship remains unresolved. Pentecostal perspectives are underrepresented; the distinctive Pentecostal contribution is a formation posture: the expectation that God actively speaks in each encounter (Chan, 1998). That posture lives in the Pastor style and formation openness axis.

Evidentiary: The analogical reasoning from fitness and therapy to spiritual formation has limits. The evidence base for digital spiritual formation apps barely exists. This field is exploratory territory.

What Building This Requires: A Leadership Reflection

Whether Aperto achieves its vision depends not only on technology and theology but on the person building it, and on the community that holds him accountable.

Mentoring UX and product design students at CareerFoundry taught me something about formation no theology textbook did. The difference between a student with curriculum and one with curriculum plus a mentor who knows their journey is categorical. Students with identical coursework produced radically different outcomes depending on whether someone noticed when they were stuck, asked "I see what you're trying to do; have you considered this?". Someonw who met them where they were and held them to higher standards. Those who struggled most didn't lack talent but someone who knew their story well enough to offer the right challenge or encouragement at the right moment.

That observation is the seed of Aperto's AI companion concept.

Co-founding *digitale christen* during the pandemic taught me something different. The initiative never launched, but navigating startup ideation with experienced founders gave me a bias toward action and pragmatism that academic theology doesn't teach. I carry those instincts into Aperto: test early, learn fast, let evidence guide the next step.

Open Source Translation as Stewardship: Making the Aperto Bible open source under Creative Commons is a conviction, not a strategy. Kairos University's tradition of capstone projects as "gifts to the church" names this well.

Financial Stewardship: The subscription model protects accessibility (free tier for core content) and integrity (refusing both ad-supported models and donation dependency). User data never sold or monetized. Formation over engagement, even when it reduces lifetime value.

Closing

I don't know if Aperto will work. The technical architecture could fail. The market assumptions could prove wrong. The theological wager on AI companionship could be a category error.

But the questions themselves aren't going away. How do you design for formation instead of engagement? How do you honor two thousand years of tradition while building with tools that didn't exist two years ago? How do you serve the people consumer religion has written off?

Those questions belong to the field now, not to any single product.

I ride the unpredictable and crowded Berlin S-Bahn and U-Bahn often. Tomorrow morning someone will pull out her phone and open something: a prayer app, meditation timer, or podcast. She won't find what this paper describes. Whether she ever does depends on whether anyone is stubborn enough to build toward formation when the market rewards engagement.

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